EECS 498/CSE 598: Zero-Knowledge Proofs
Winter 2026
Lecture 20

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Beyster 4709

Agenda for this lecture

- Announcements
- Motivating incremental proofs
- Incrementally verifiable computation (IVC) via recursion
- IVC via folding, the Nova construction
- Proof-carrying data

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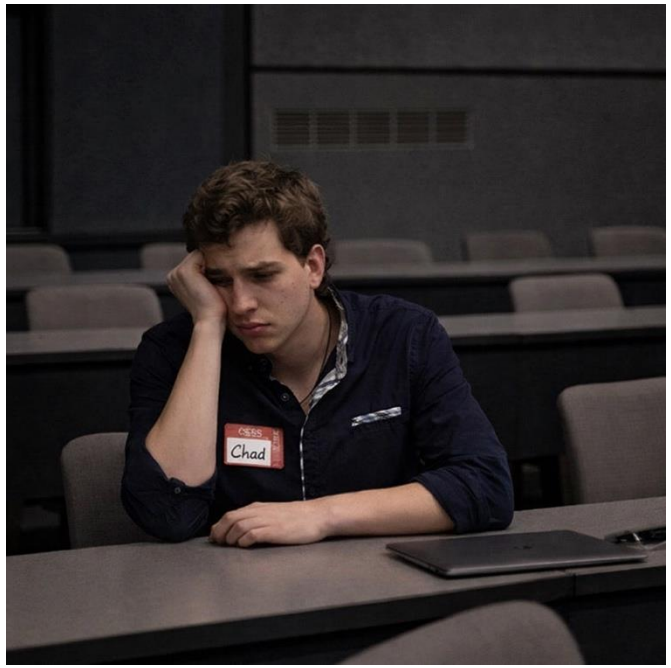
Announcements

- The plan is to release P4 and theory pset 2 ASAP.
  - Note there will be no P5 nor a final exam. These will be the last assignments of the semester.
- 598 research track: **project proposal due Friday 3/27**
- All work for this course must be completed by our final exam date, April 27th
- Reminder: you can get participation credit for going to discussion.
  - Chad is getting lonely on Fridays!



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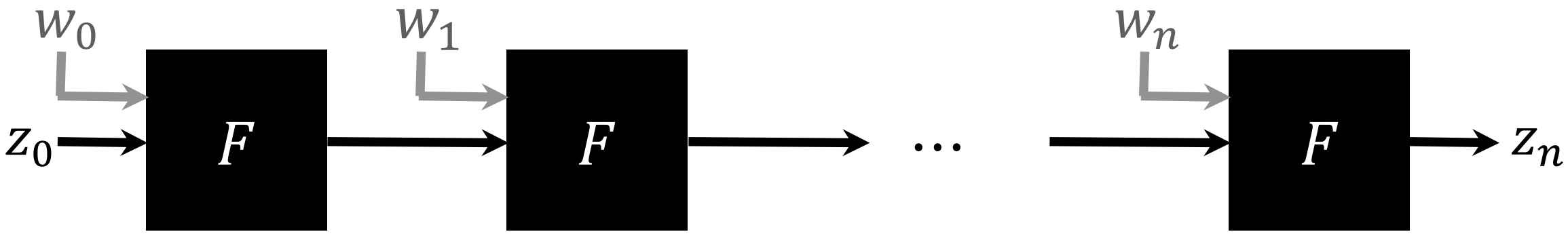
Most of the slides in this lecture are due to Abhiram Kothapalli

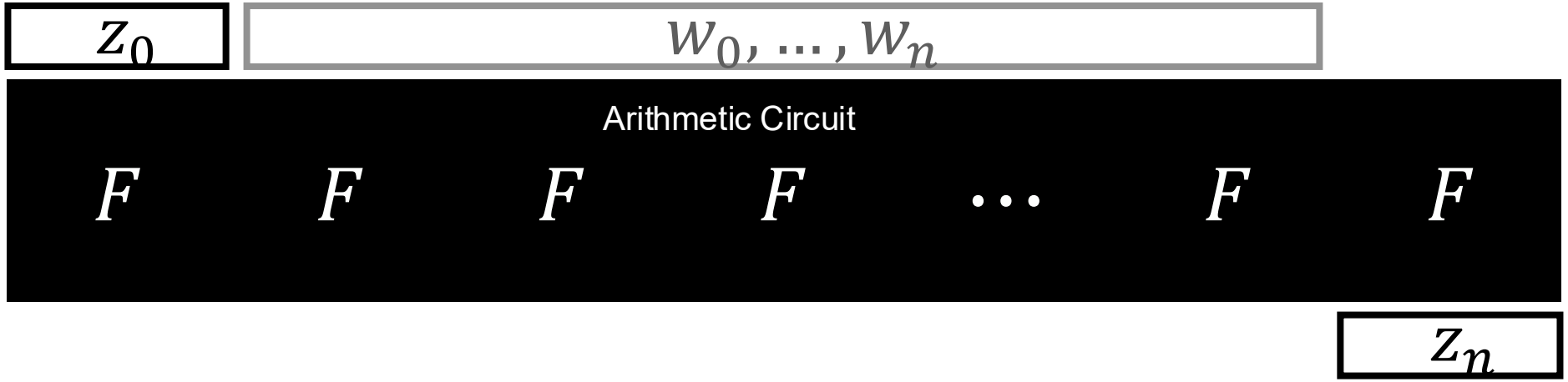
Goal: Practical SNARKs for Recursive Computation

Proving VM execution reduces to proving that function F applied n times to initial input z_0 results in z_n



Naive Approach

Use a SNARK for Circuit-SAT to monolithically prove the unrolled statement



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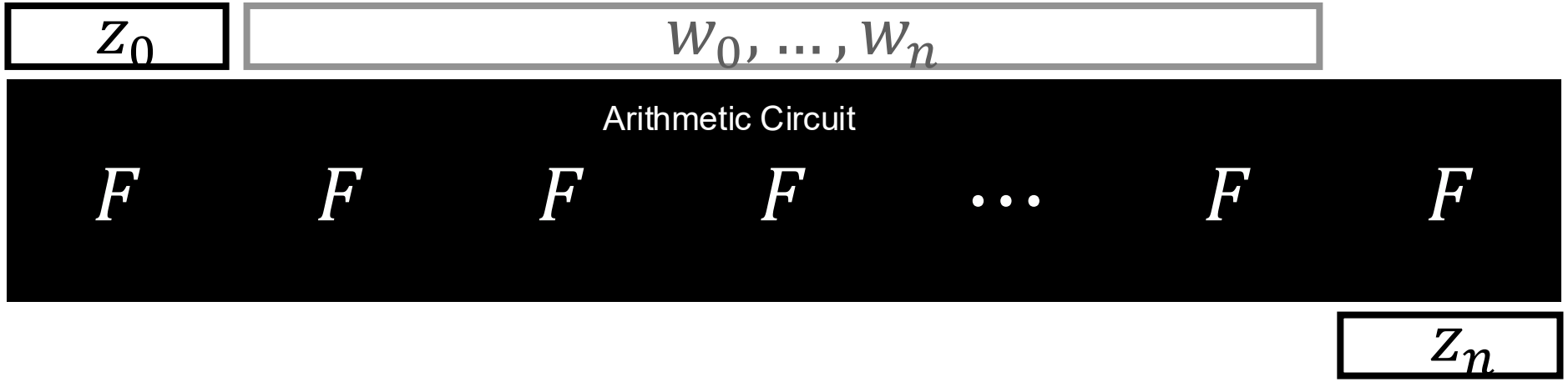

Drawbacks

- Fixes n ahead of time
- $O(n \cdot |F|)$ prover memory and verifier preprocessing time
- Verifier time may depend on n

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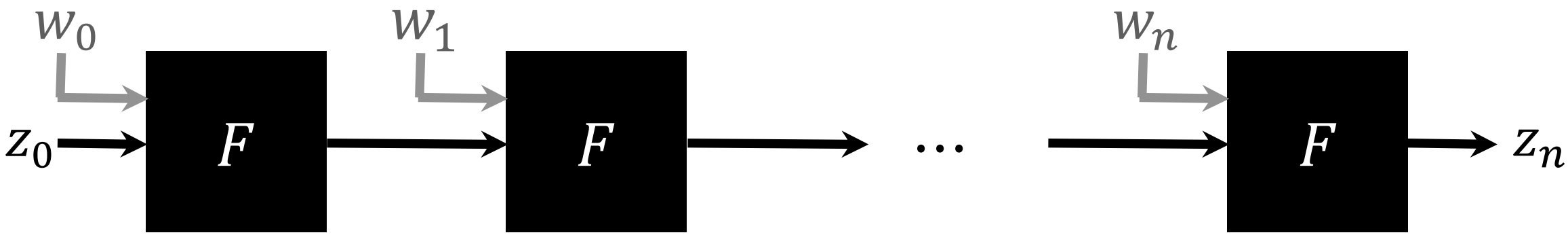
Approach: Incrementally Verifiable Computation (IVC) [Val08]

Incrementally update a proof of i applications to a proof of $i + 1$ applications with the same size



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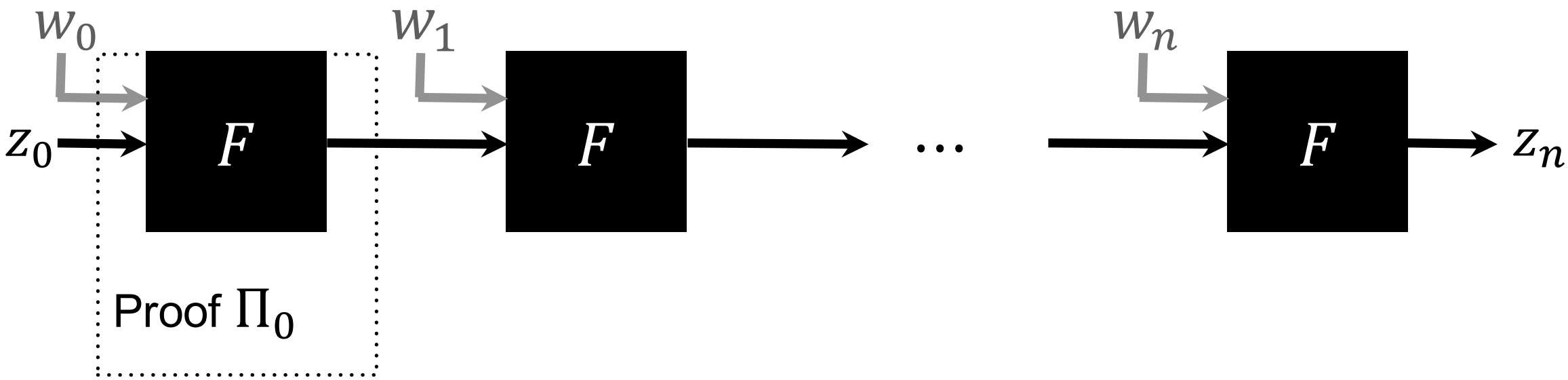

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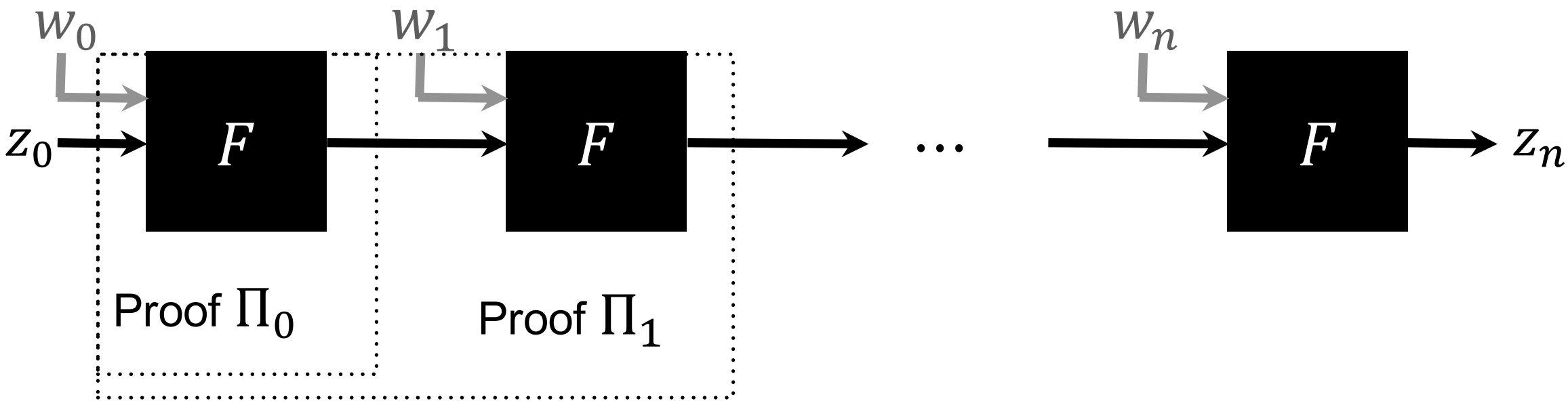

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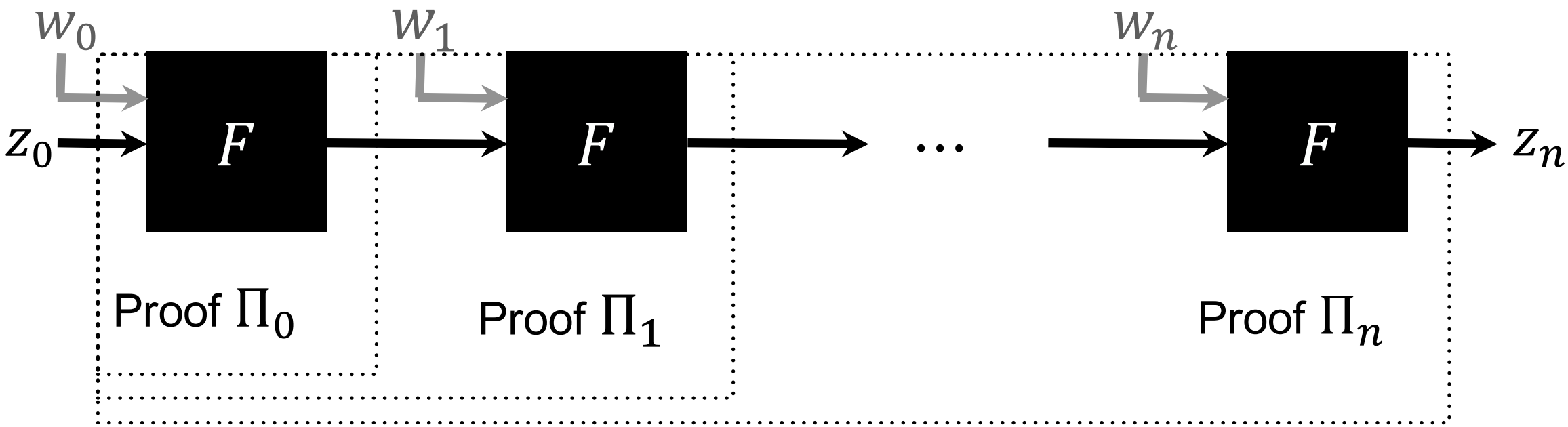
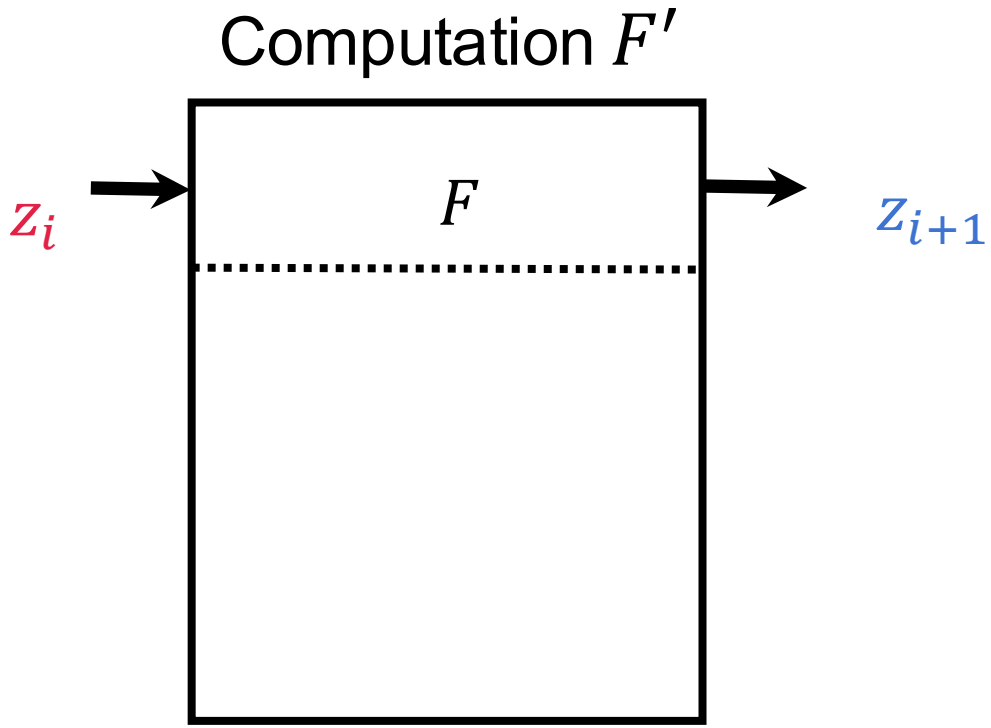


Valiant's Original IVC Approach [Val08,BCTV14]



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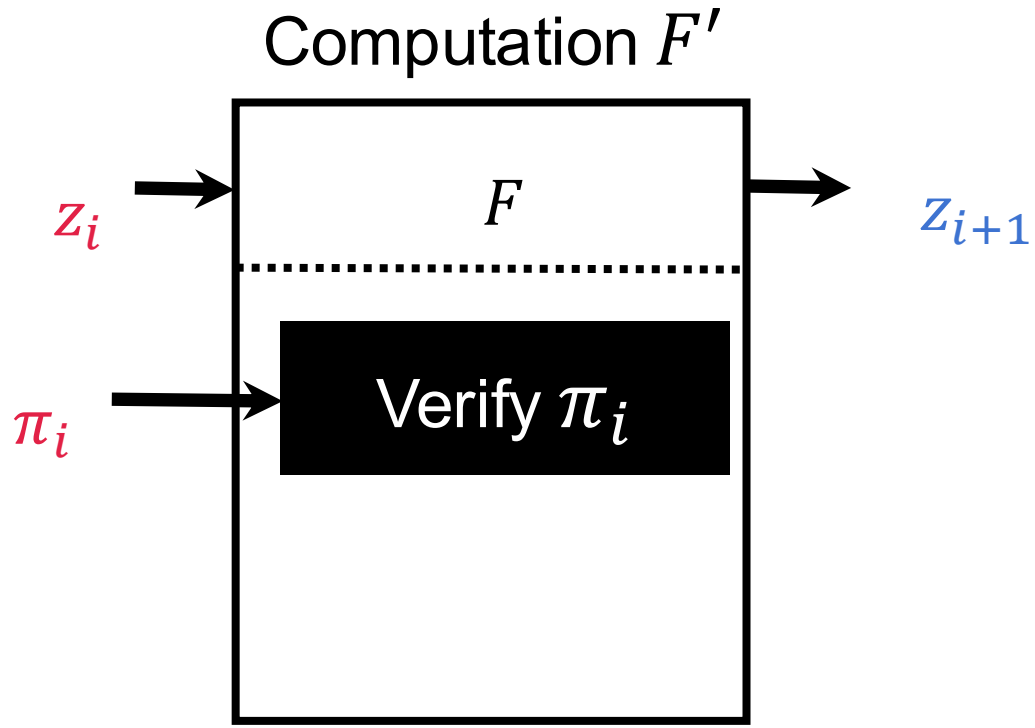

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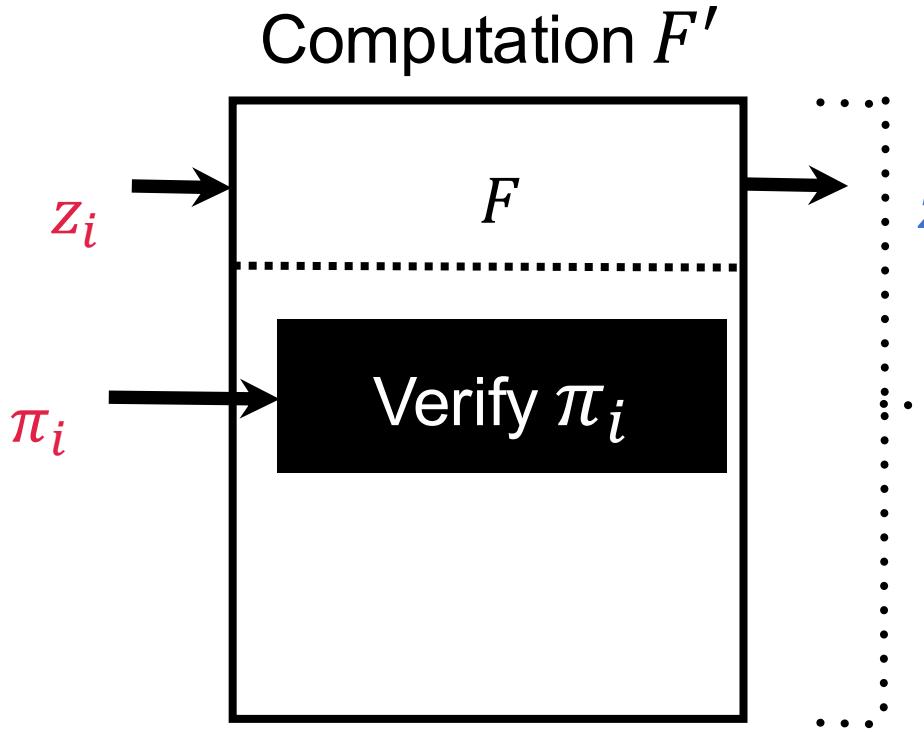

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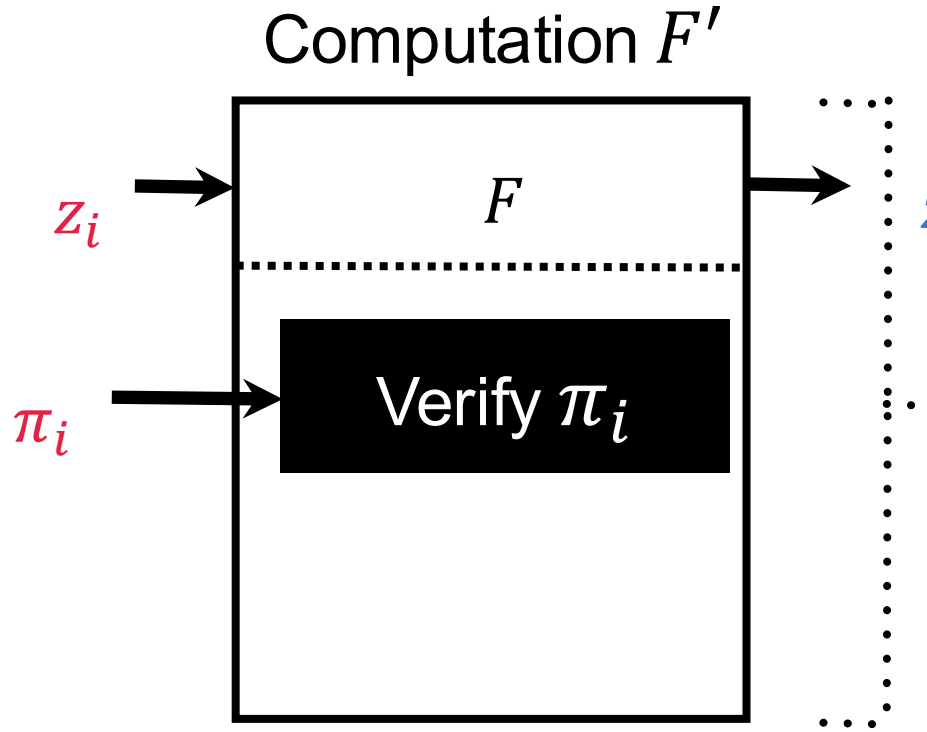


SNARK Proof of $(F')^i(z_0) = z_i$

SNARK Proof of $(F')^{i+1}(z_0) = z_{i+1}$

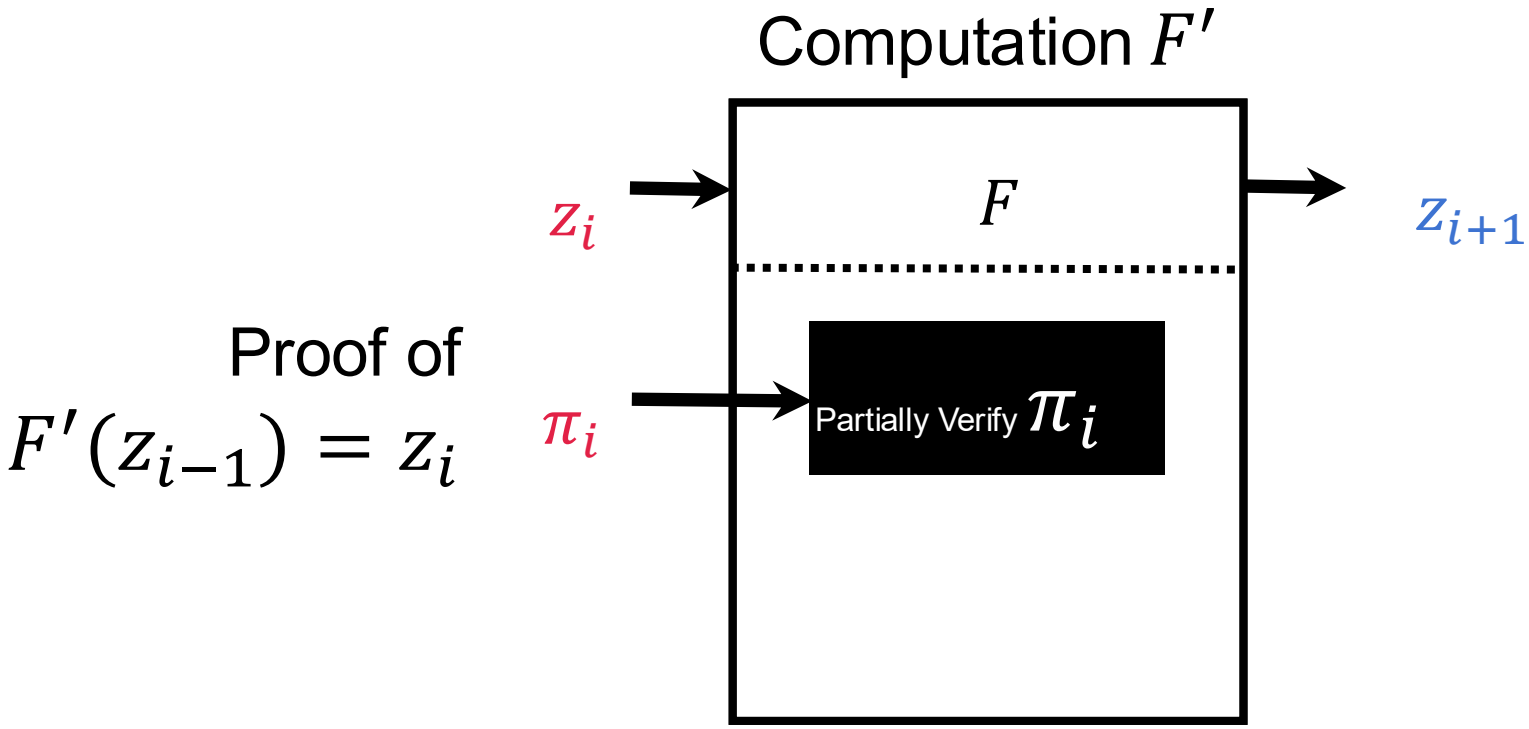
$|E/\mathbb{F}_p| = q$

$|E/\mathbb{F}_q| = p$

Drawbacks

- Trusted-setup SNARKs require expensive cycles of pairing-friendly curves
- Utilizing SNARKs without trusted setup require much larger verifier circuits

Halo's Approach: Partially Verify Proofs [BGH19,BCLMS20,BDFG20]



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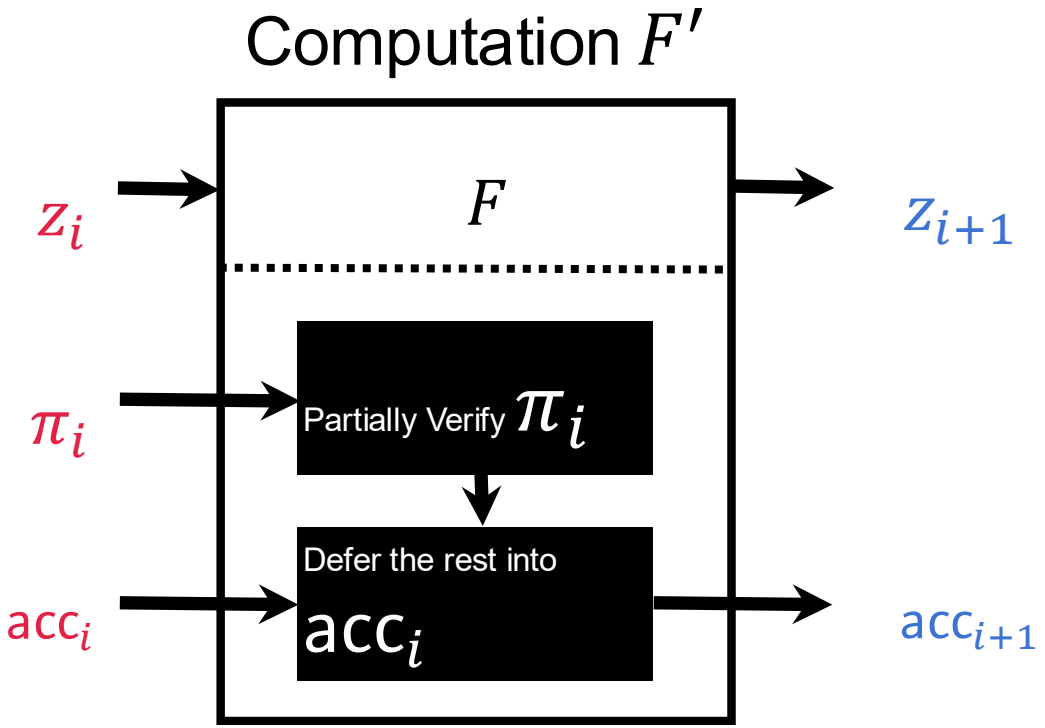

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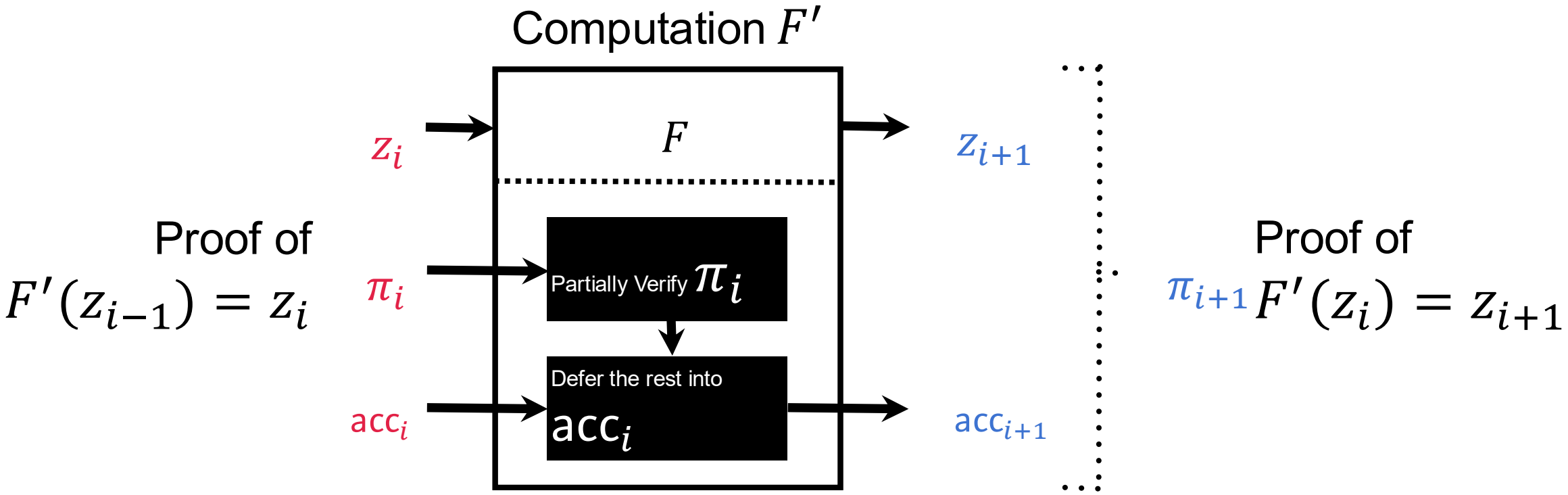

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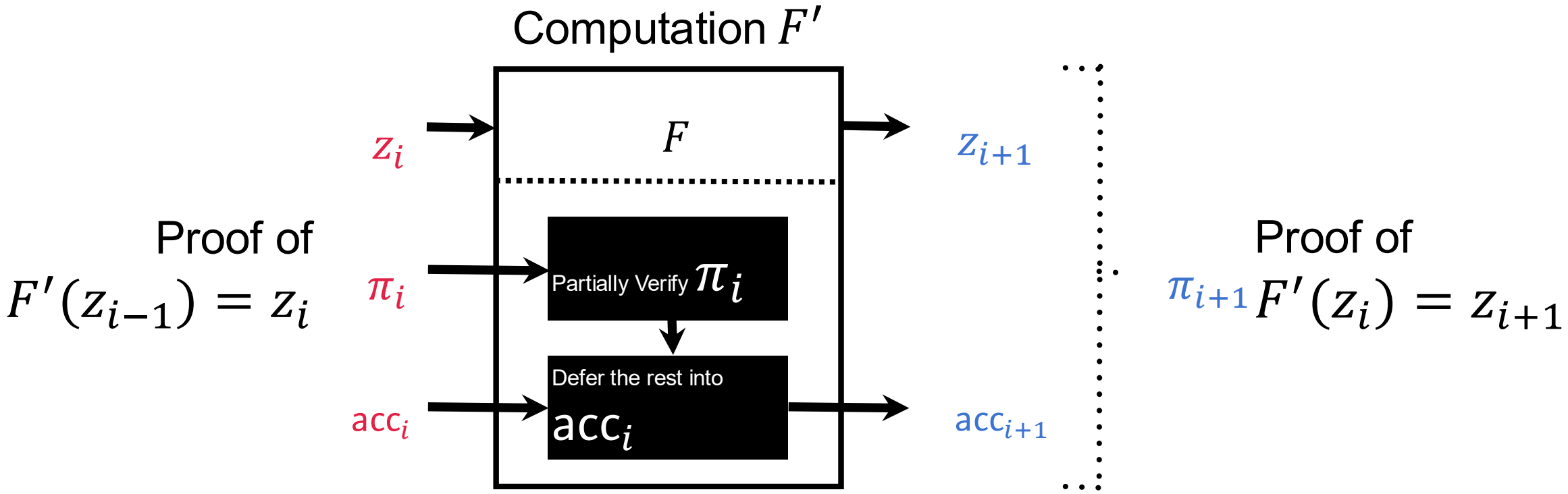
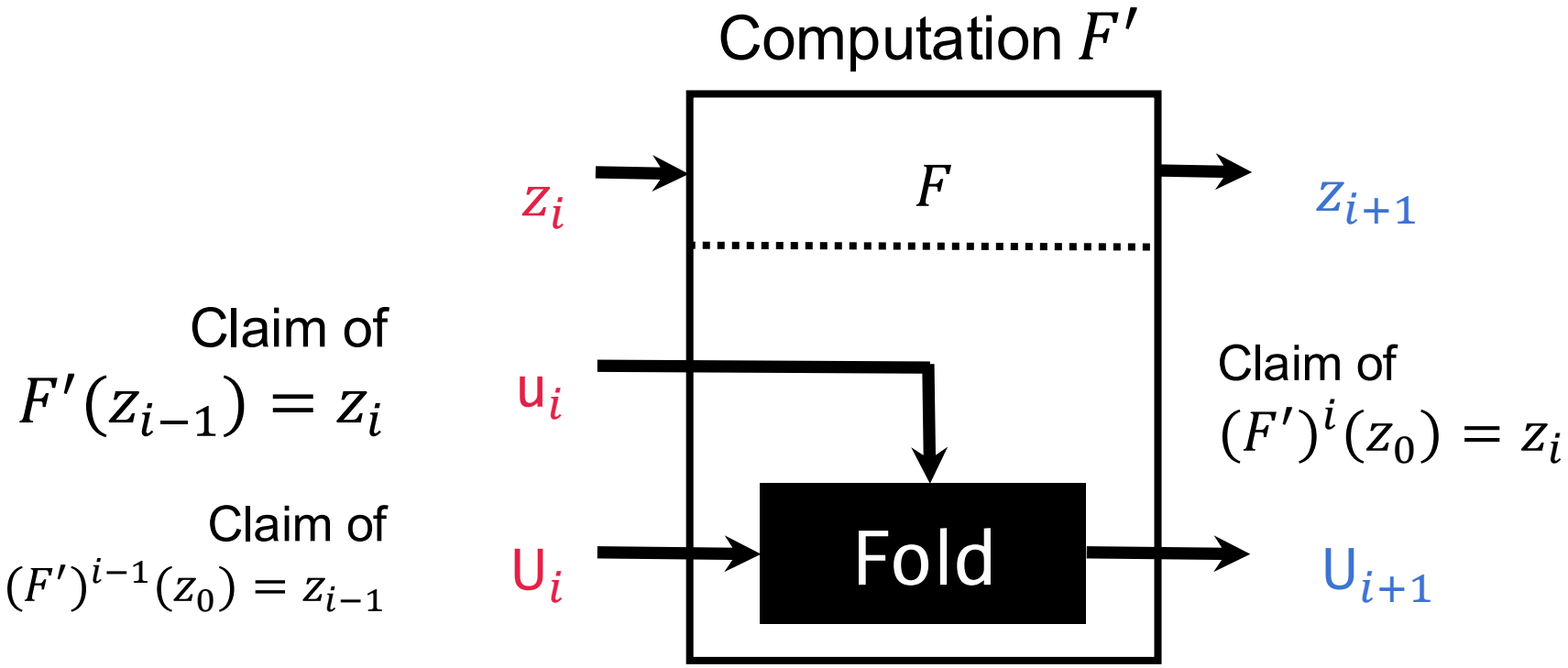


Drawbacks

- Partially checking π_i in-circuit is still expensive
- Generating π_i is concretely and asymptotically expensive

Nova: Reduce Claims rather than Verify Proofs



Fold reduces the task of checking two claims to the task of checking a single claim of the same size

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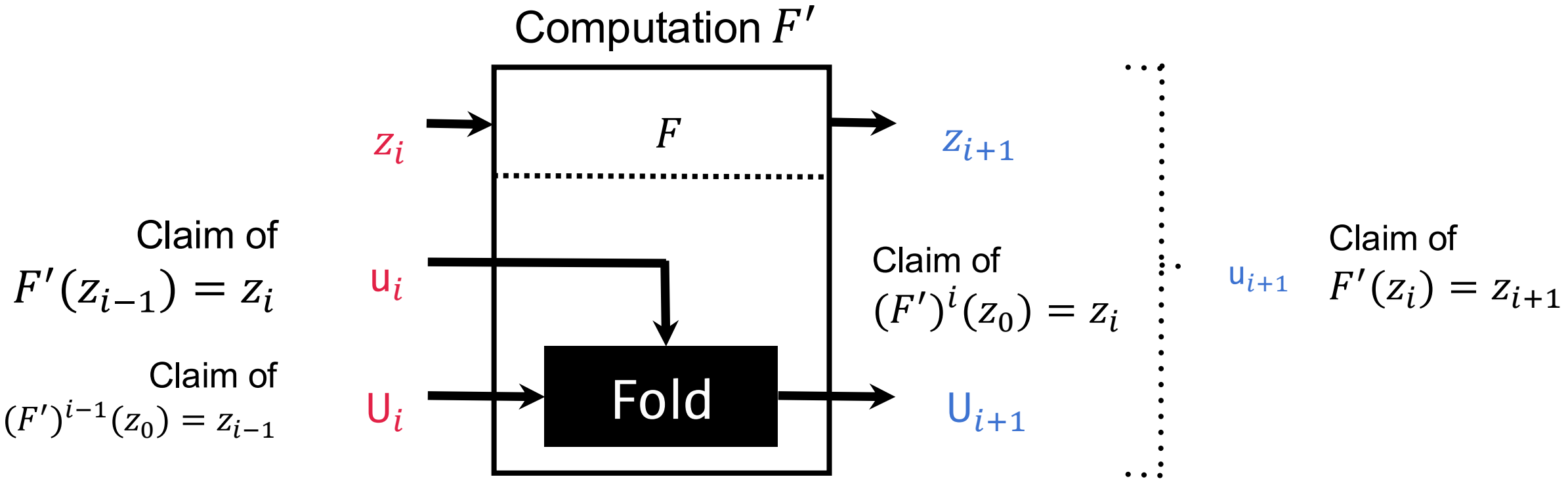

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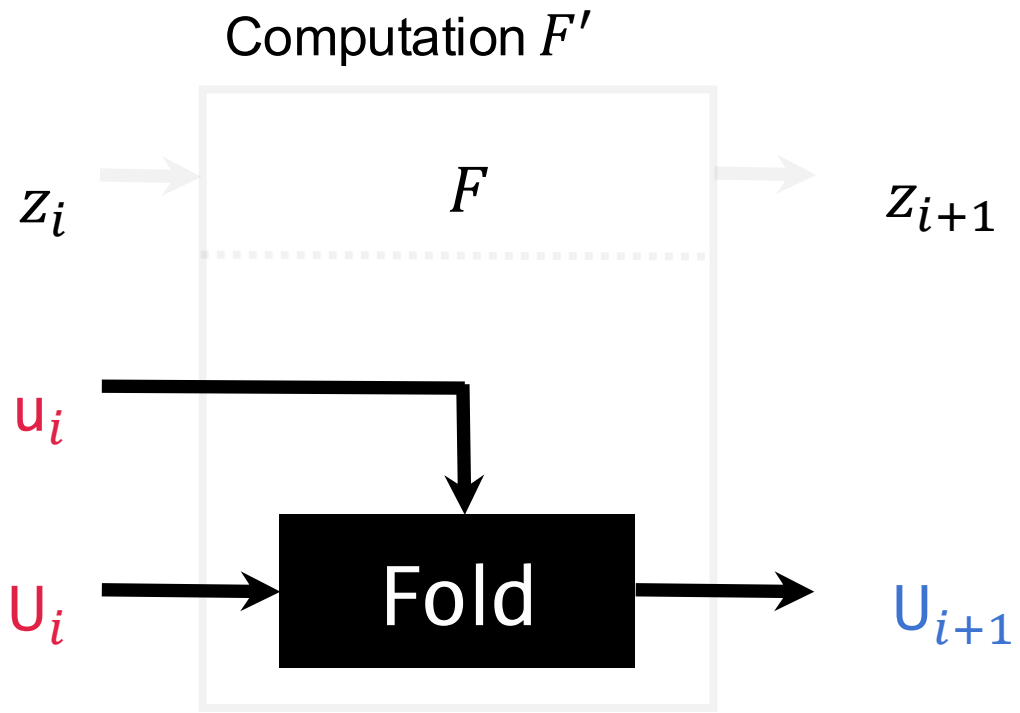
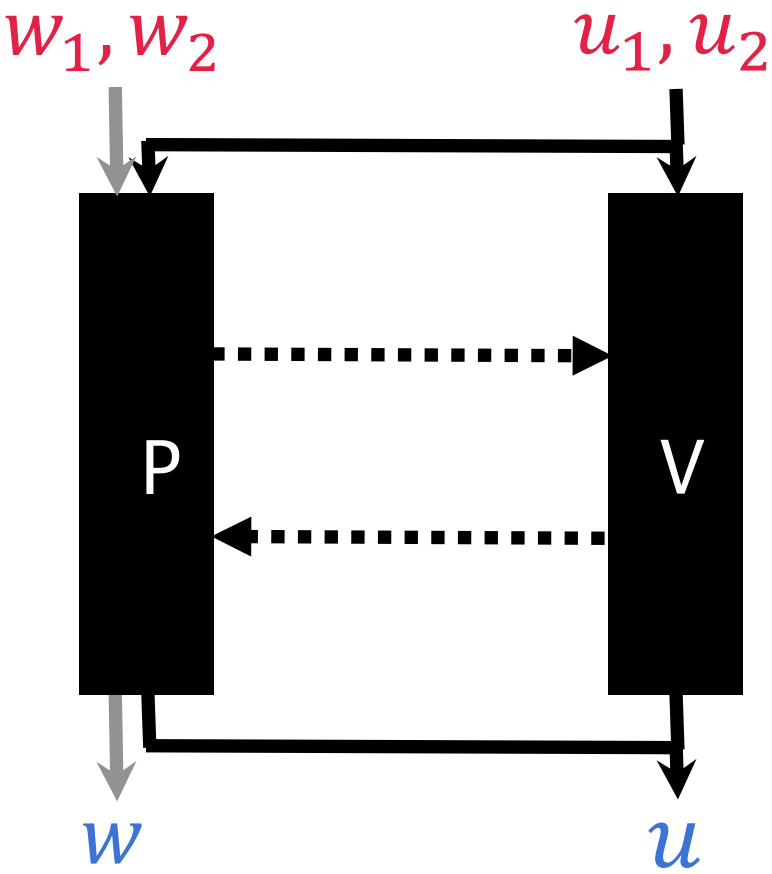
How do we implement *Fold*?



Relaxed Requirement: Interactive Folding Scheme

A folding scheme interactively reduces the claims $(u_1, w_1), (u_2, w_2) \in R$ to a claim $(u, w) \in R$



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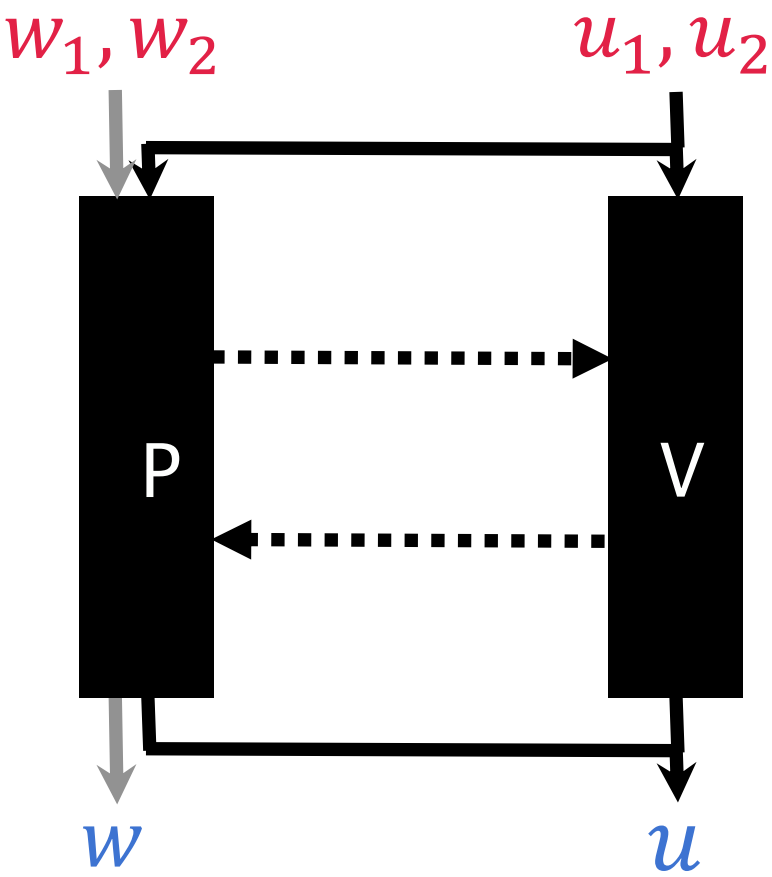

Completeness

If u_1, u_2 are satisfiable then u is satisfiable

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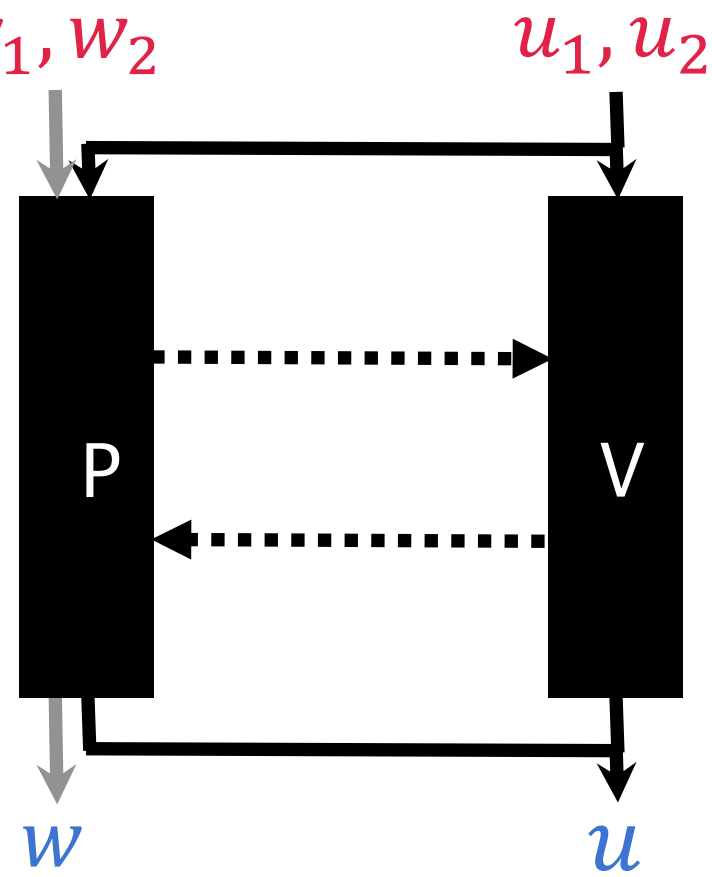

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If prover outputs satisfying w then it must know satisfying w_1, w_2

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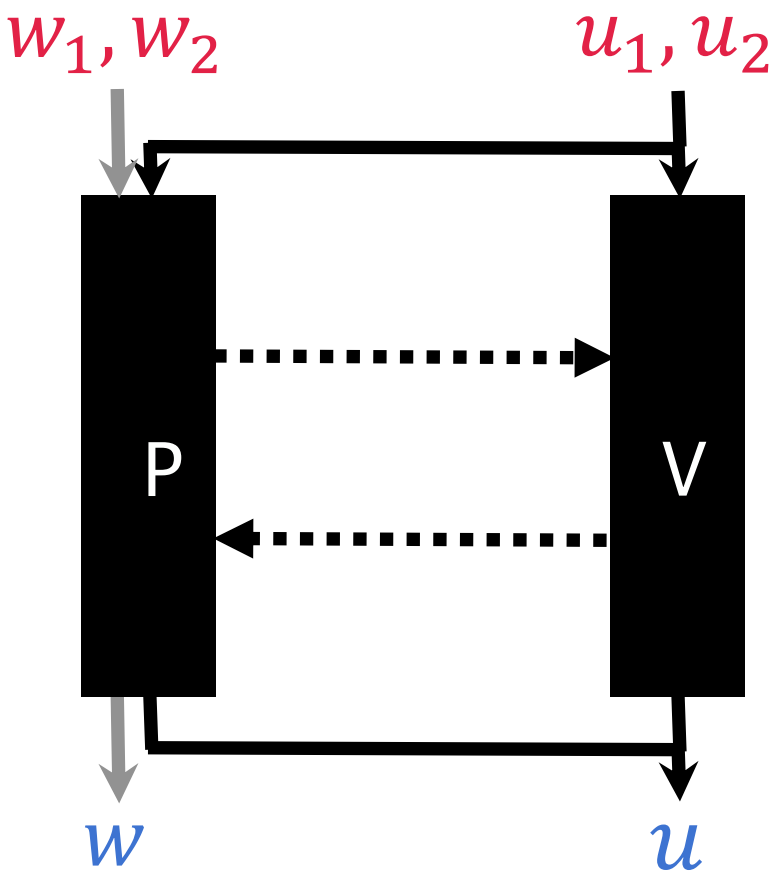


Completeness

If u_1, u_2 are satisfiable then u is satisfiable

Knowledge Soundness

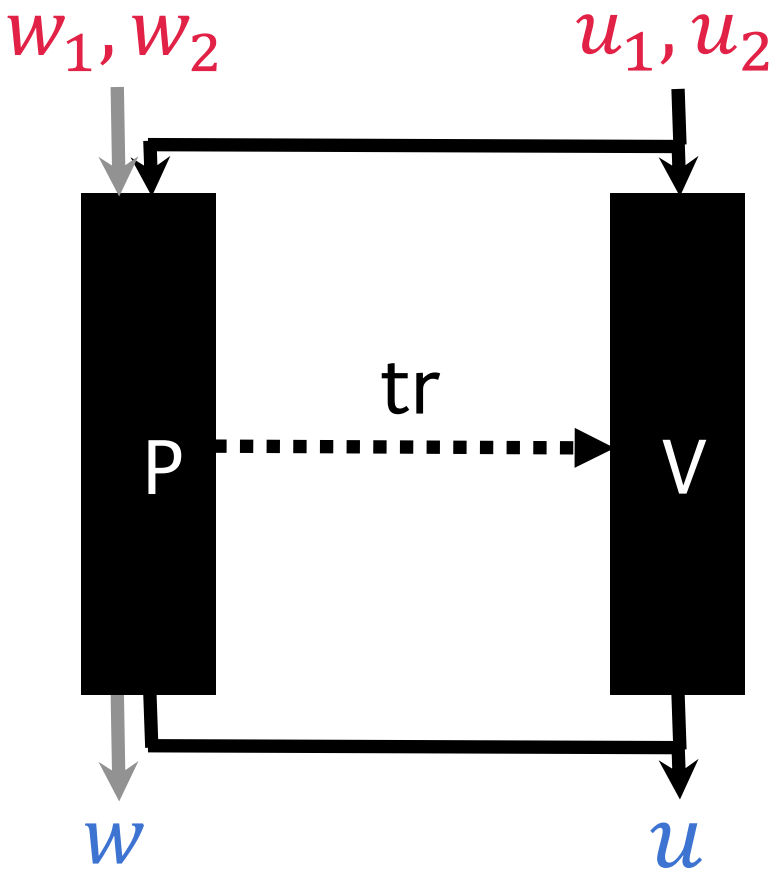
If prover outputs satisfying w then it must know satisfying w_1, w_2

Succinctness

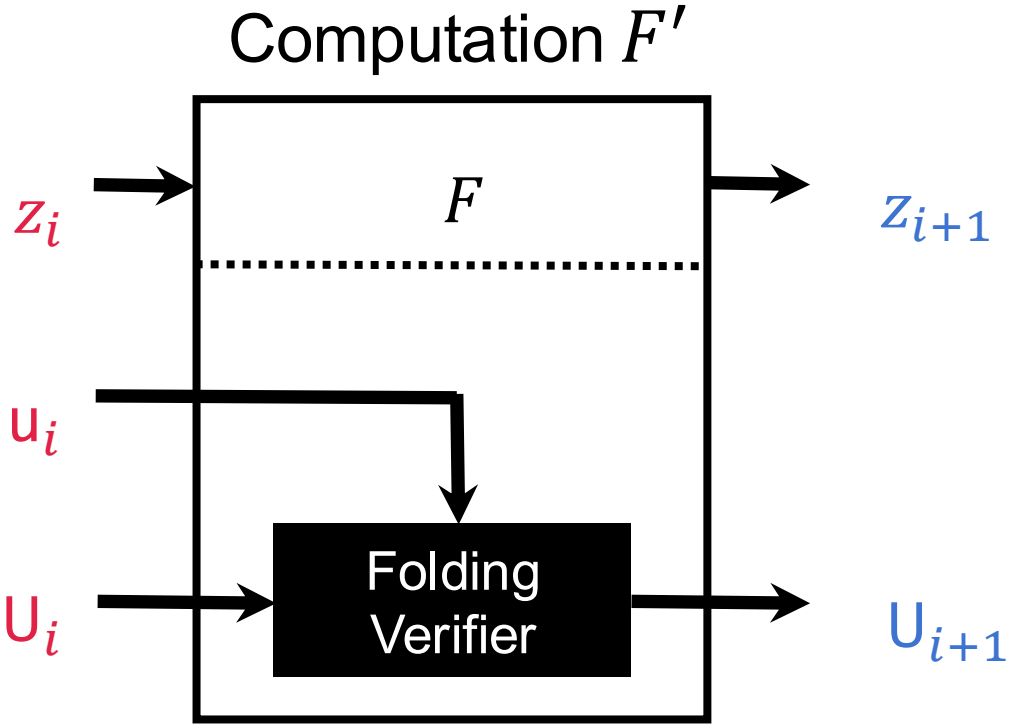
Folding should be more efficient than checking an instance

Non-Interactive Folding Schemes

A public-coin folding scheme can be made non-interactive via the Fiat-Shamir Transform.



Given a non-interactive folding scheme for NP, F' can verifiably fold u_i and U_i by running the folding verifier.



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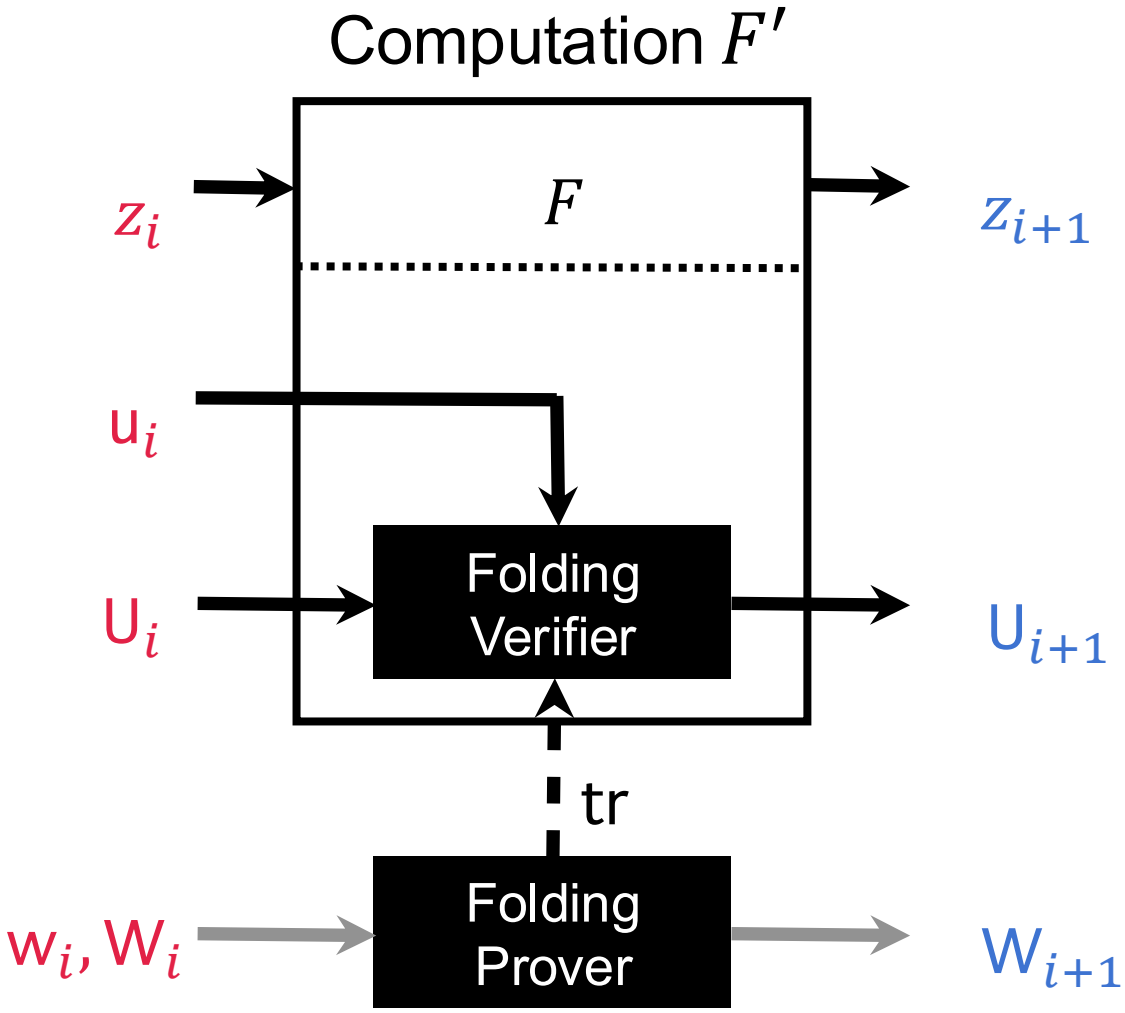

Witness w_i consists of a trace of the execution of $F(z_{i-1}) = z_i$

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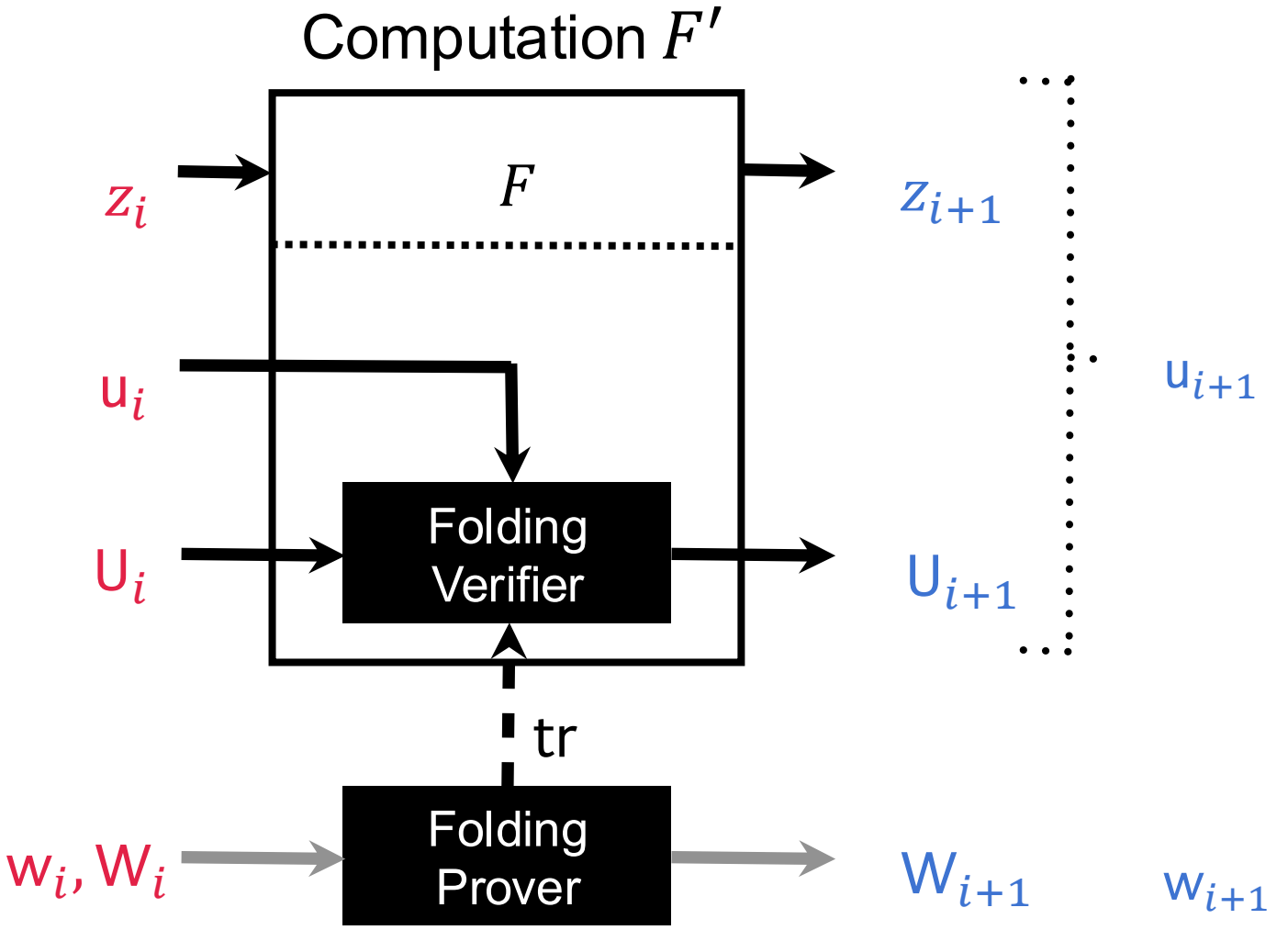


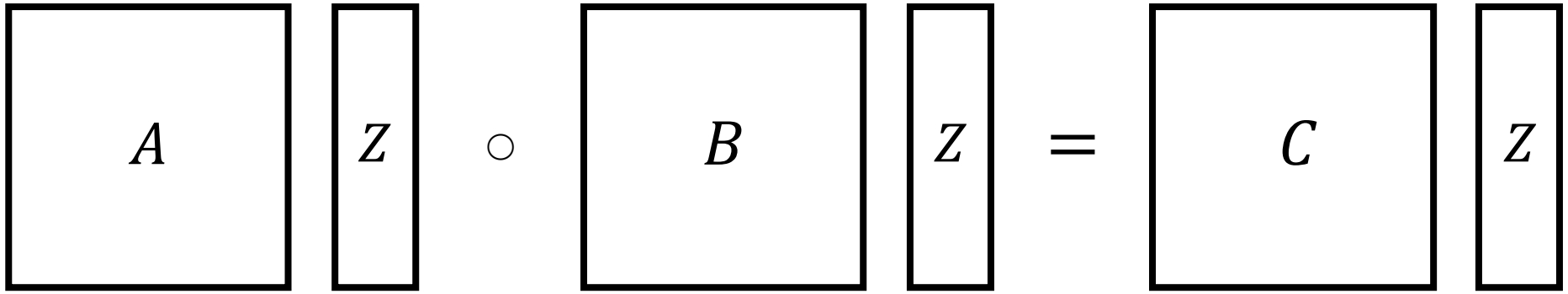
Claim of $F'(z_i) = z_{i+1}$

Witness w_i consists of a trace of the execution of $F(z_{i-1}) = z_i$

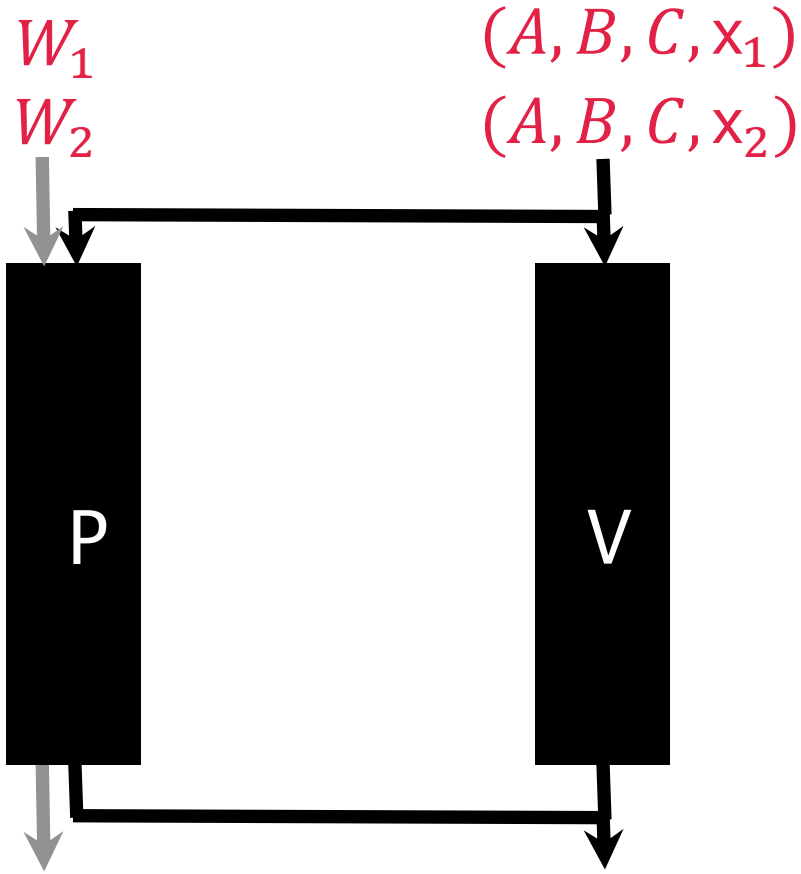
Folding an NP-Complete Relation

We start with R1CS

An R1CS statement consists of constraint matrices A, B, C, and vector x. A witness vector W is satisfying if for $Z = (W, \mathrm{x}, 1)$



Attempt to Fold R1CS Instances



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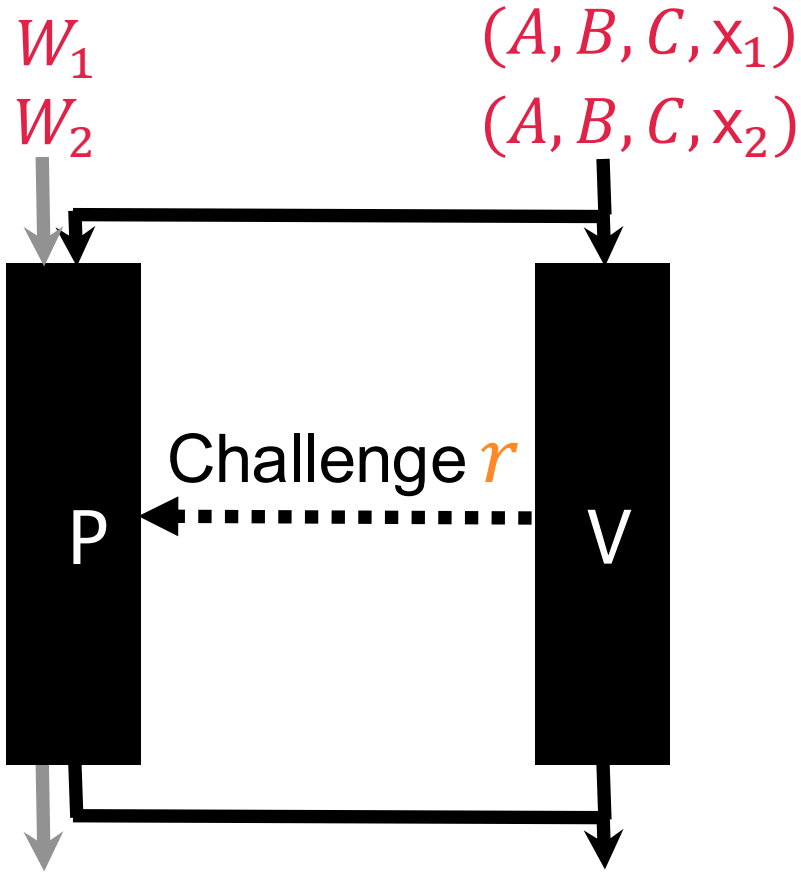

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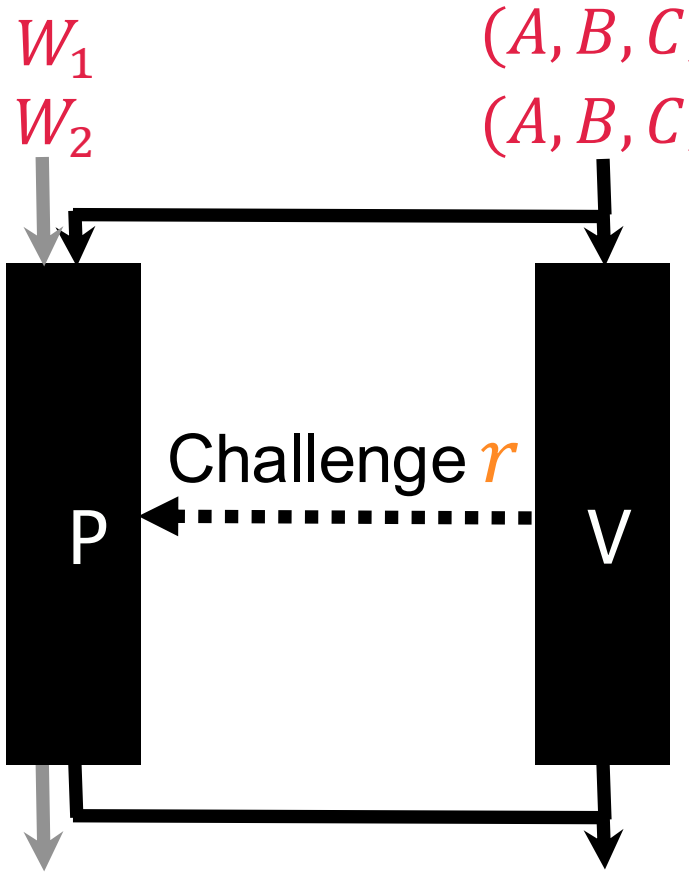

$$\mathrm{x} \leftarrow \mathrm{x}_1 + r \cdot \mathrm{x}_2$$

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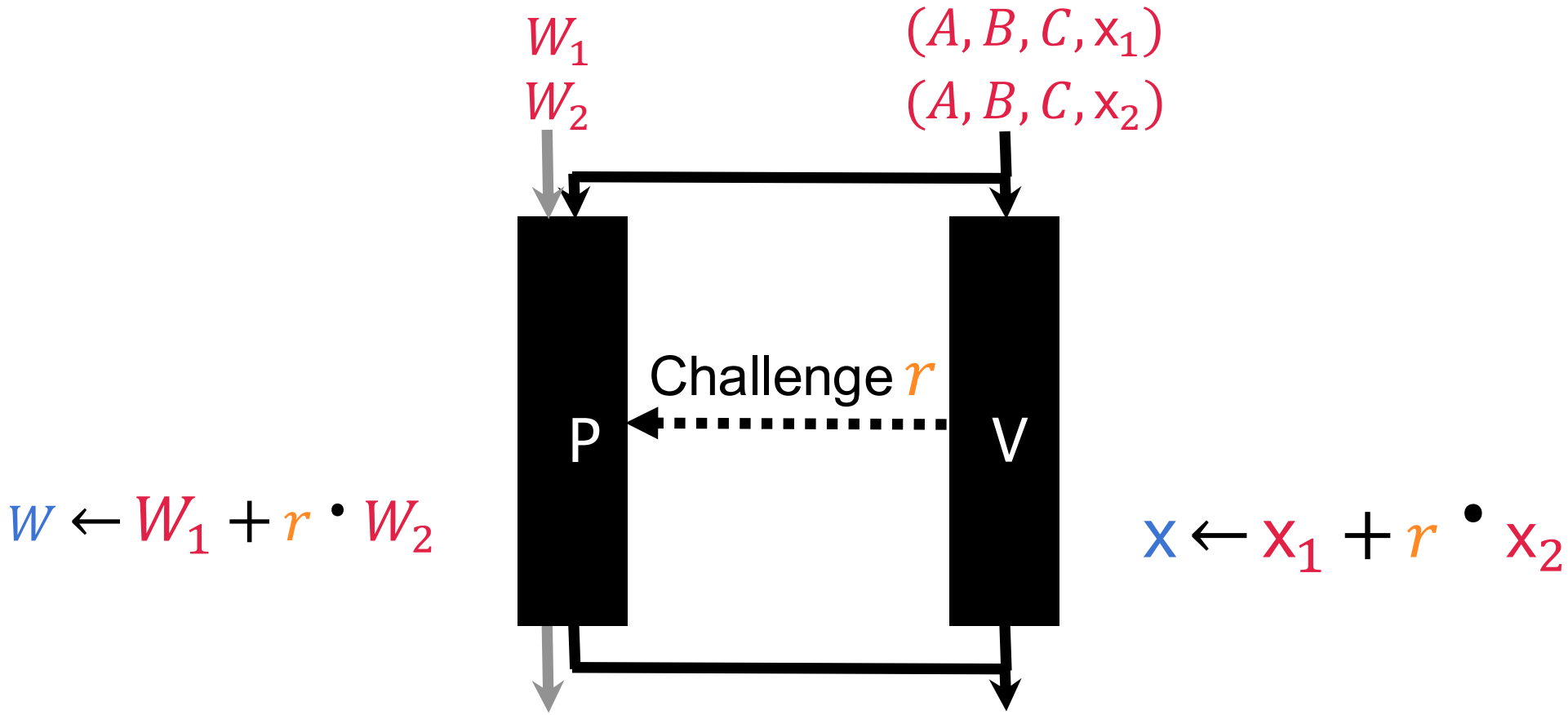

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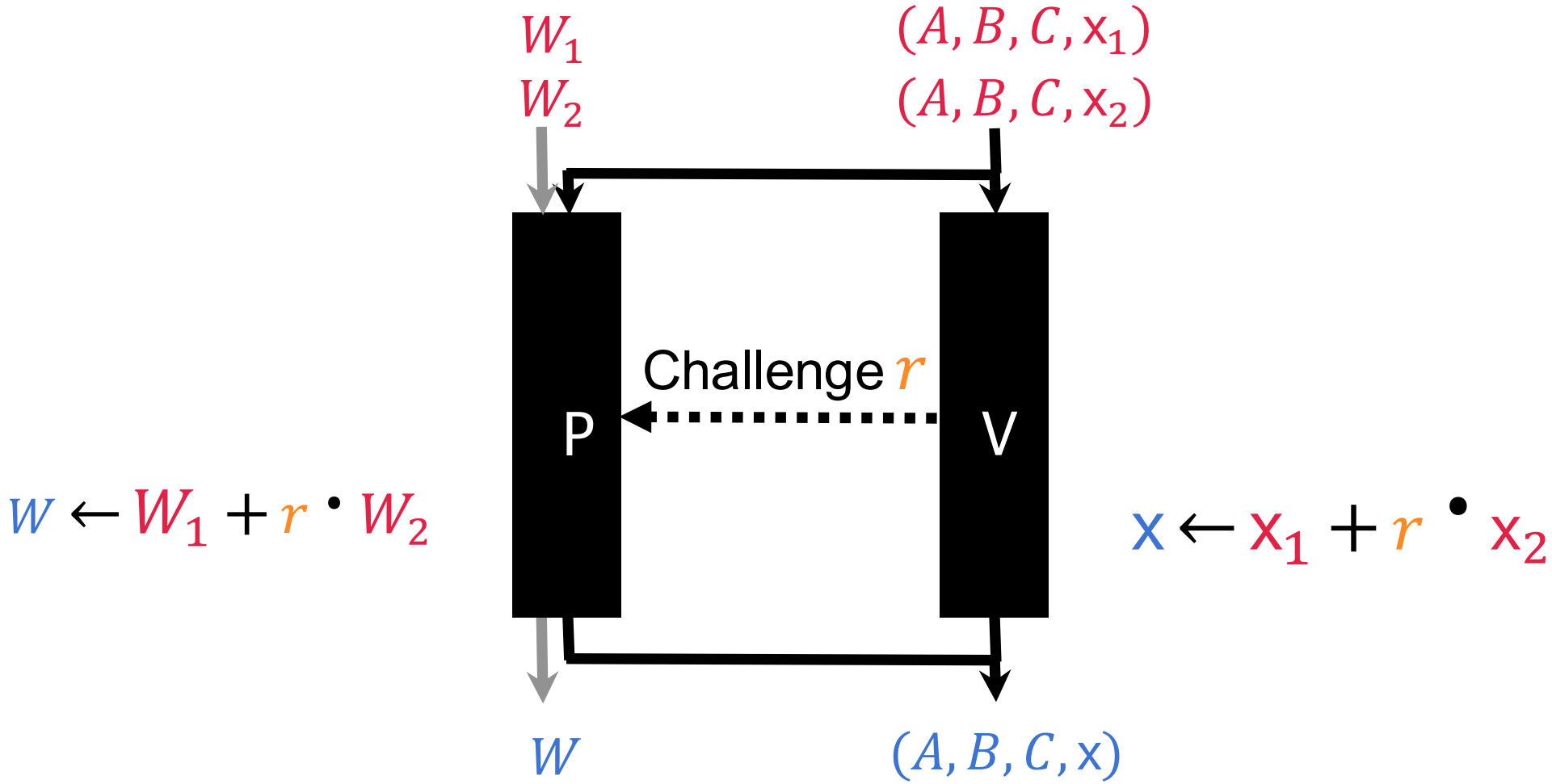

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$$AZ \circ BZ = AZ_1 \circ BZ_1 + r \cdot (AZ_1 \circ BZ_2 + AZ_2 \circ BZ_1) + r^2 \cdot AZ_2 \circ BZ_2$$

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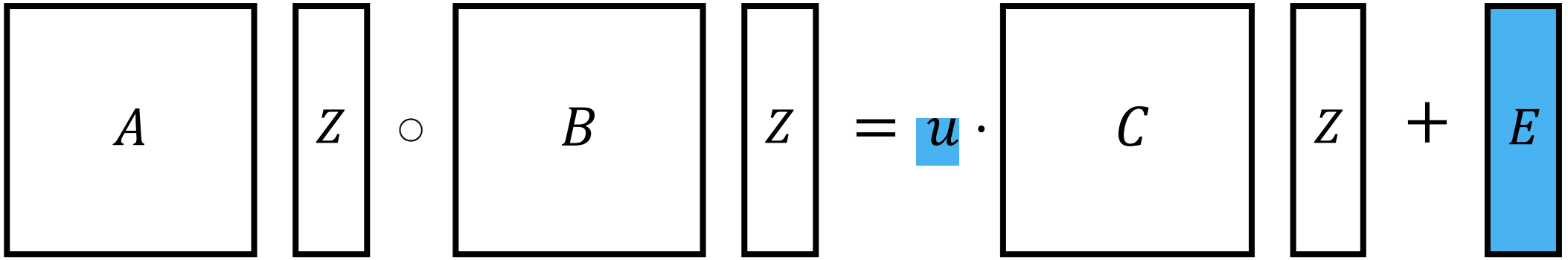
To absorb the extra r factor we introduce scalar u in the statement

$$AZ \circ BZ = AZ_1 \circ BZ_1 + r \cdot (AZ_1 \circ BZ_2 + AZ_2 \circ BZ_1) + r^2 \cdot AZ_2 \circ BZ_2$$

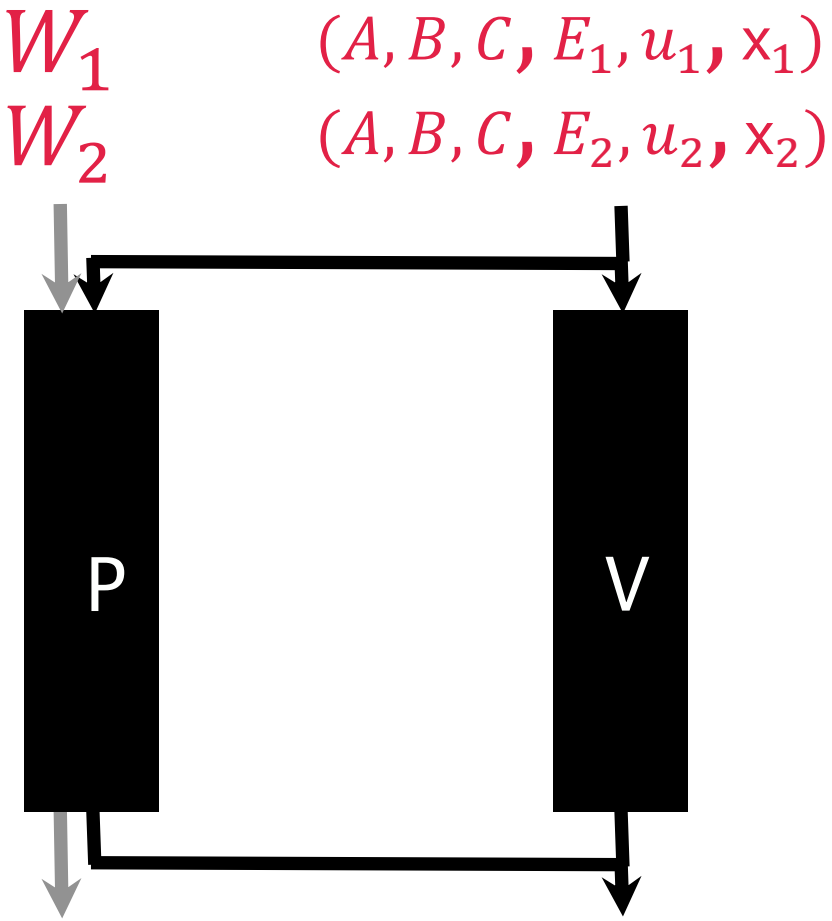
To absorb the error terms we introduce an error vector E in the statement

Relaxed R1CS: A Variant of R1CS Amenable to Folding

A relaxed R1CS statement additionally contains an error vector E and scalar u. A witness vector W is satisfying if for $Z = (W, \mathrm{x}, u)$



Folding Scheme for Relaxed R1CS (Almost)



Folding Scheme for Relaxed R1CS (Almost)

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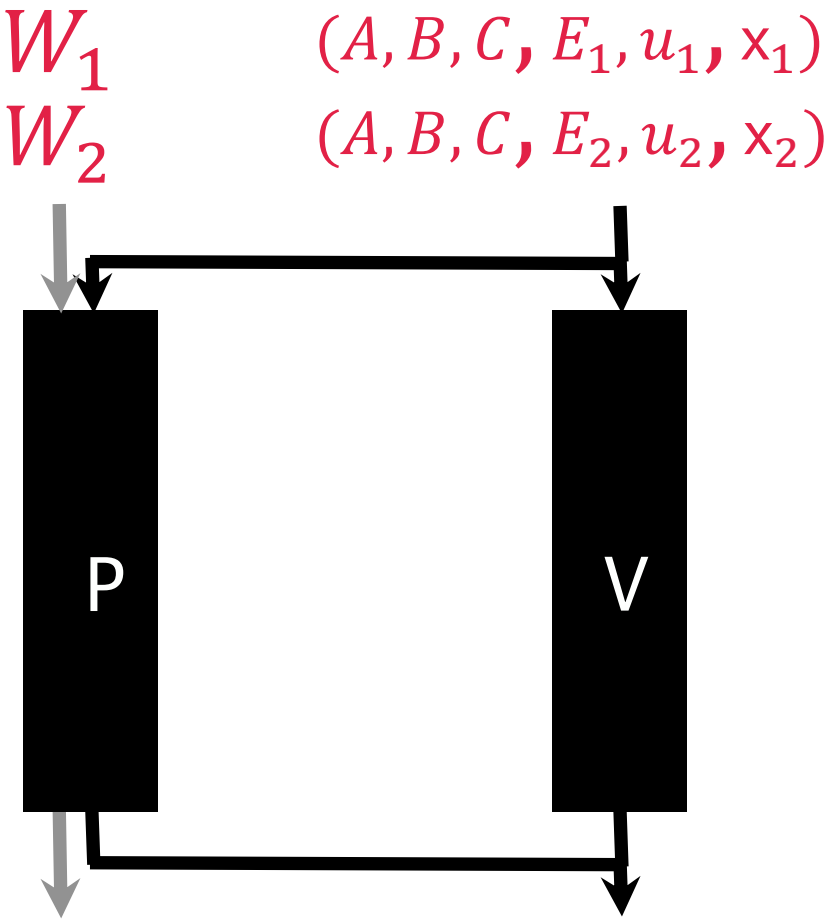

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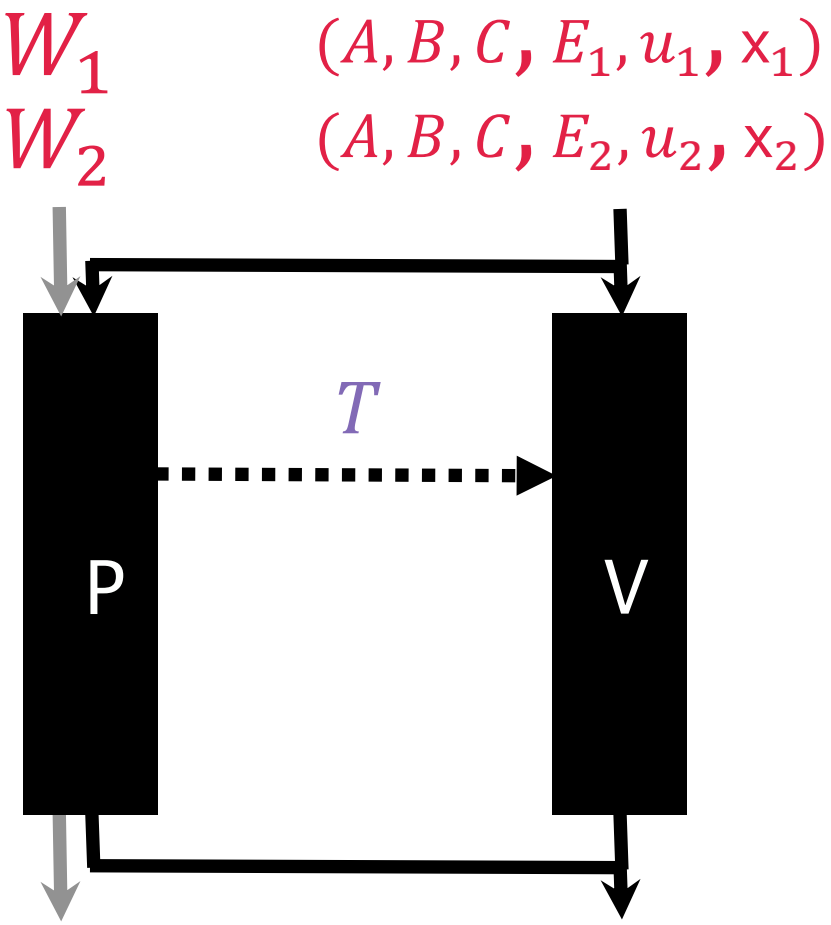

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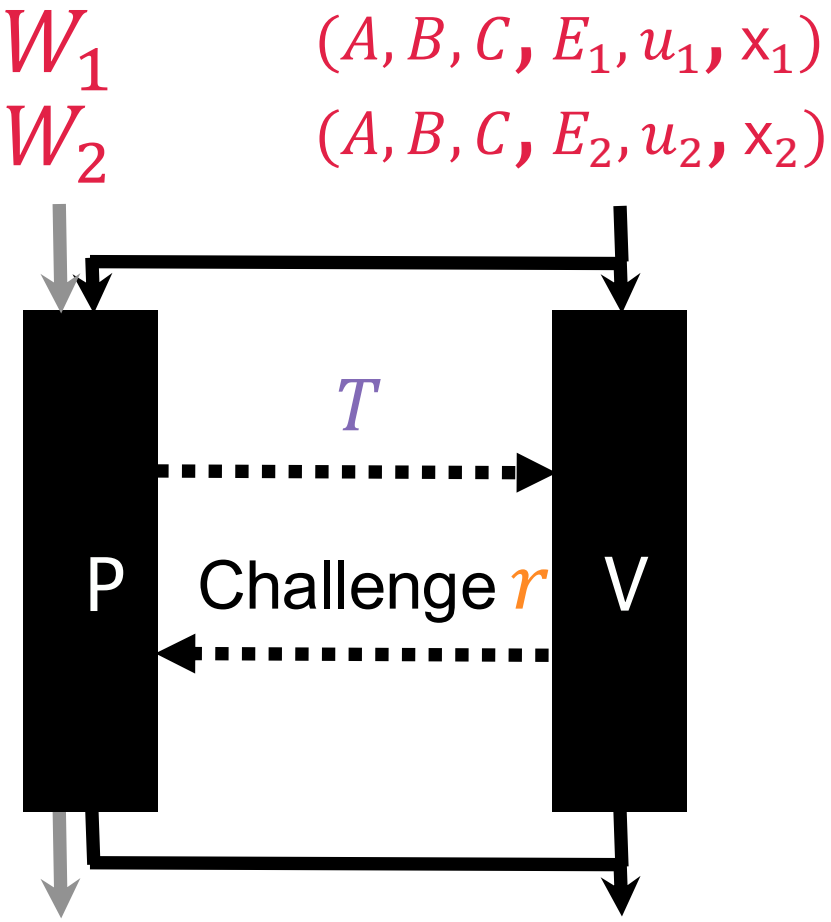
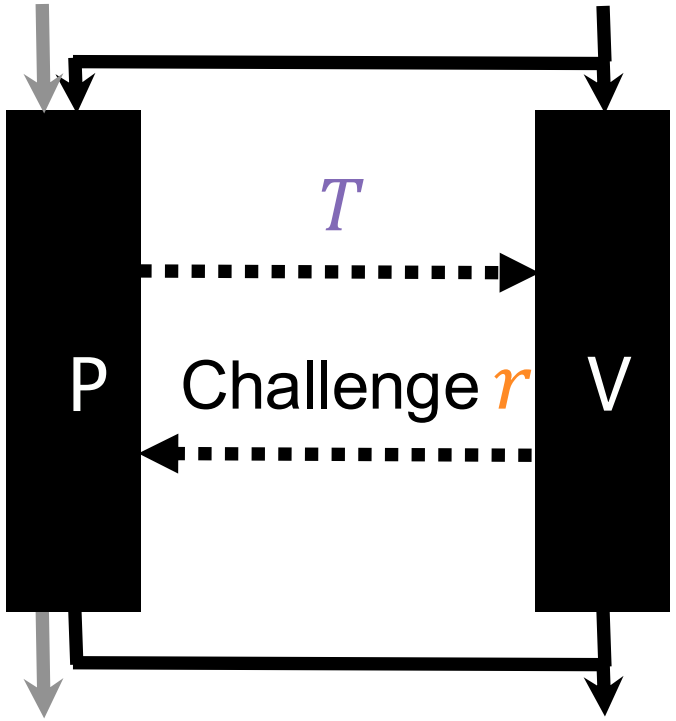


Folding Scheme for Relaxed R1CS (Almost)

W_1 $(A, B, C, E_1, u_1, \mathrm{x}_1)$

W_2 $(A, B, C, E_2, u_2, \mathrm{x}_2)$

$T \leftarrow AZ_1 \circ BZ_2 + AZ_2 \circ BZ_1 - u_1 \cdot CZ_2 - u_2 \cdot CZ_1$



$u \leftarrow u_1 + r \cdot u_2$

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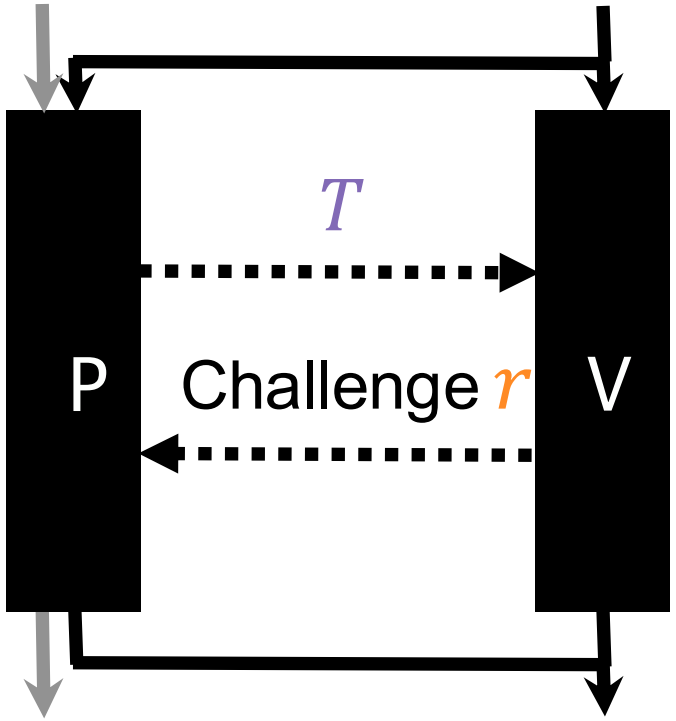

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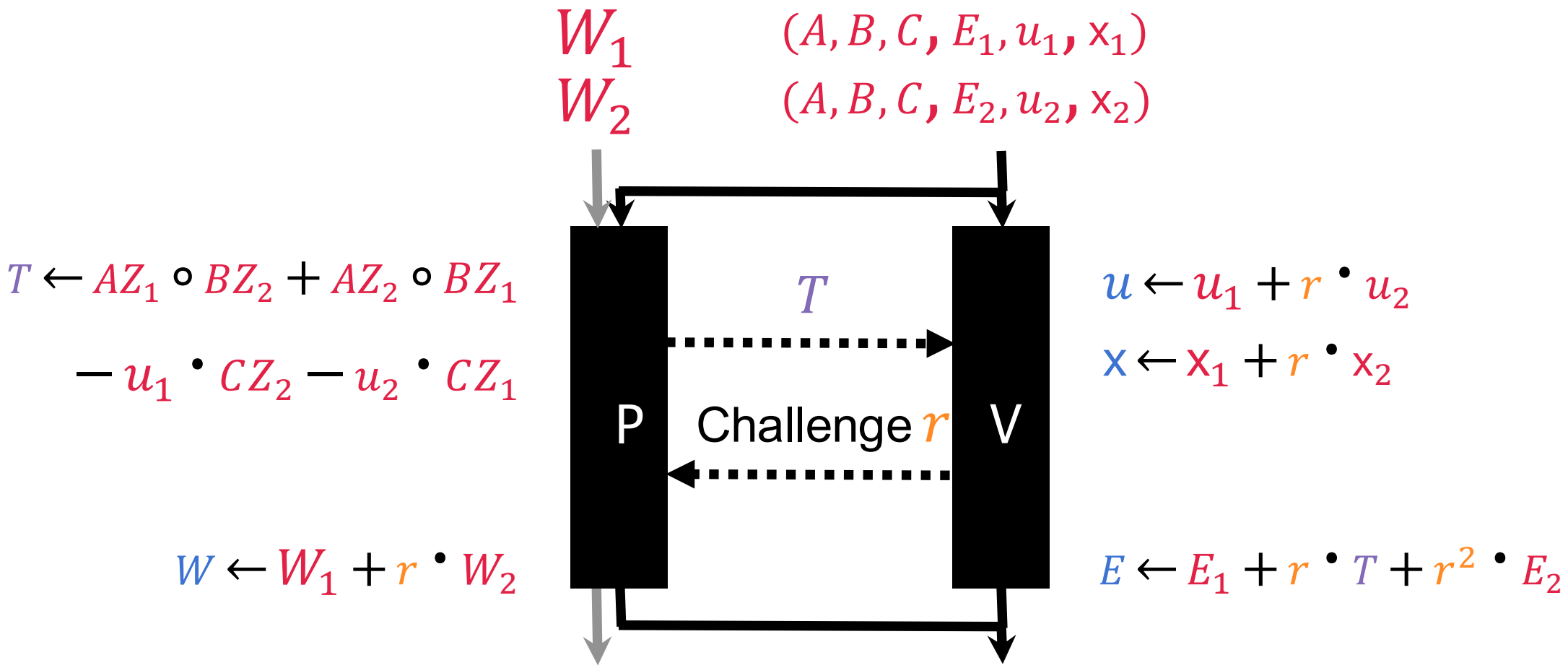

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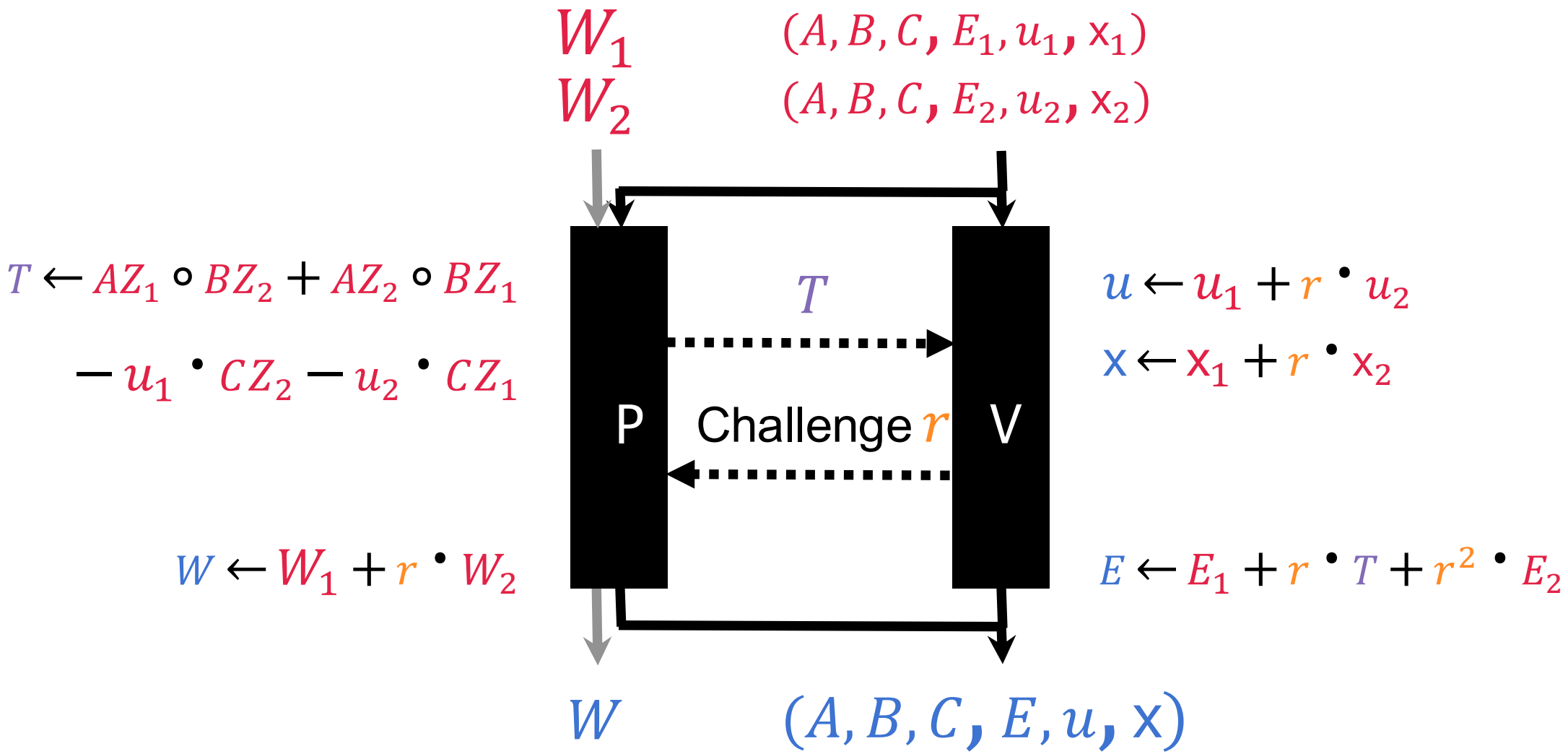

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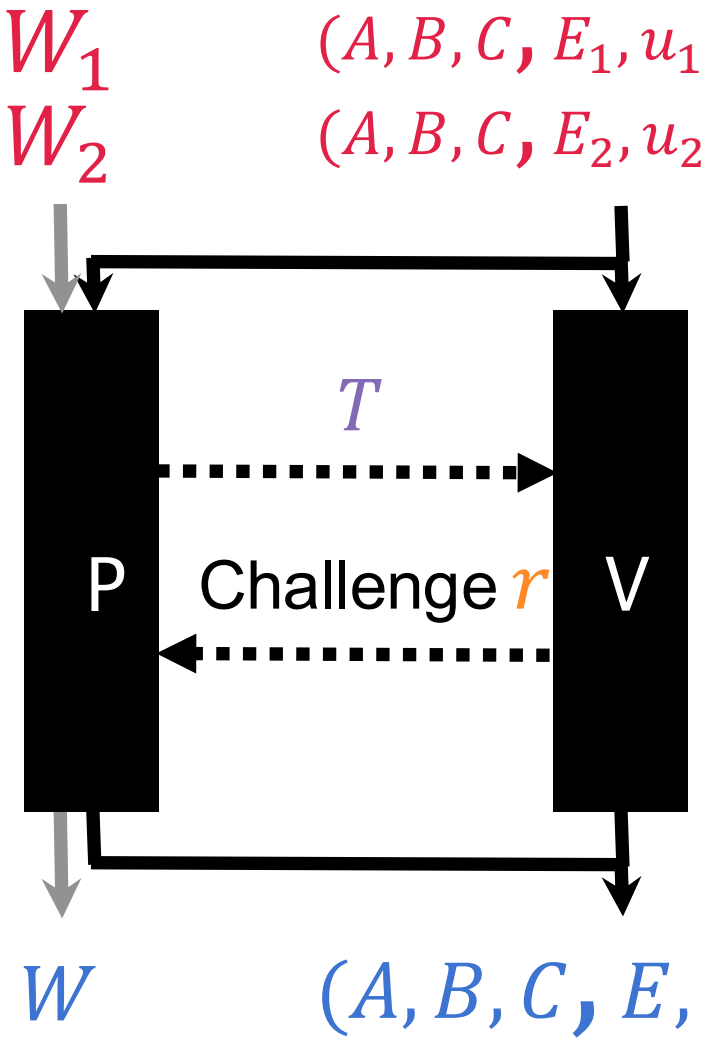
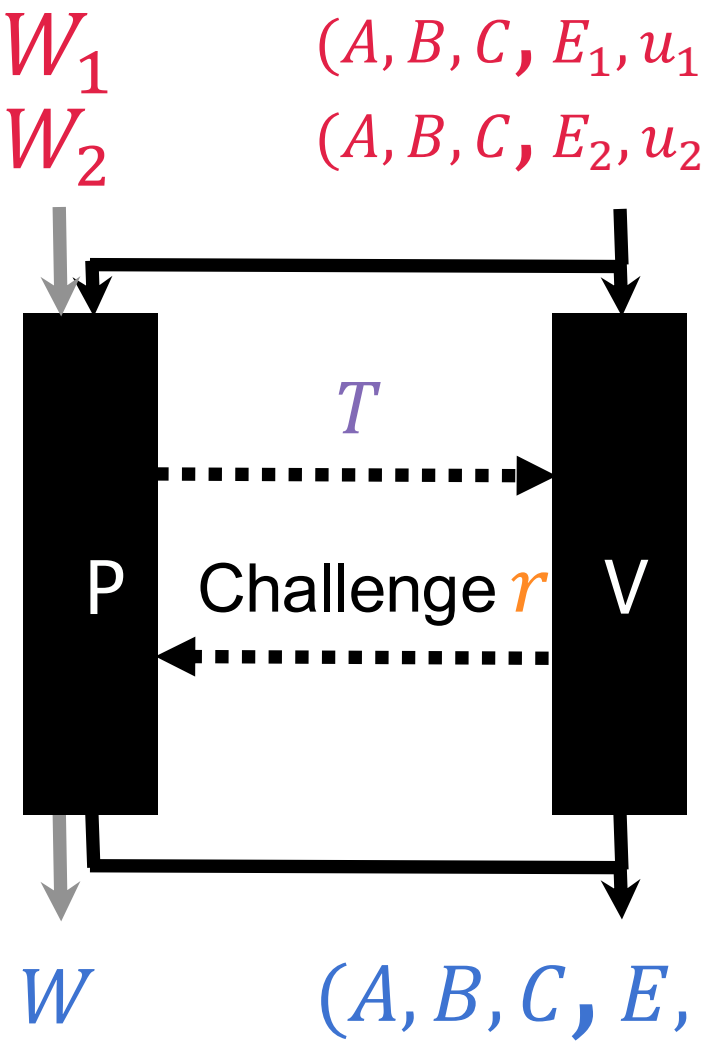


Problem: Verifier cannot enforce that prover folds W correctly

Folding Scheme for Relaxed R1CS (Almost)

Problem:
Because $|E| = O(|W|)$ the verifier is not succinct



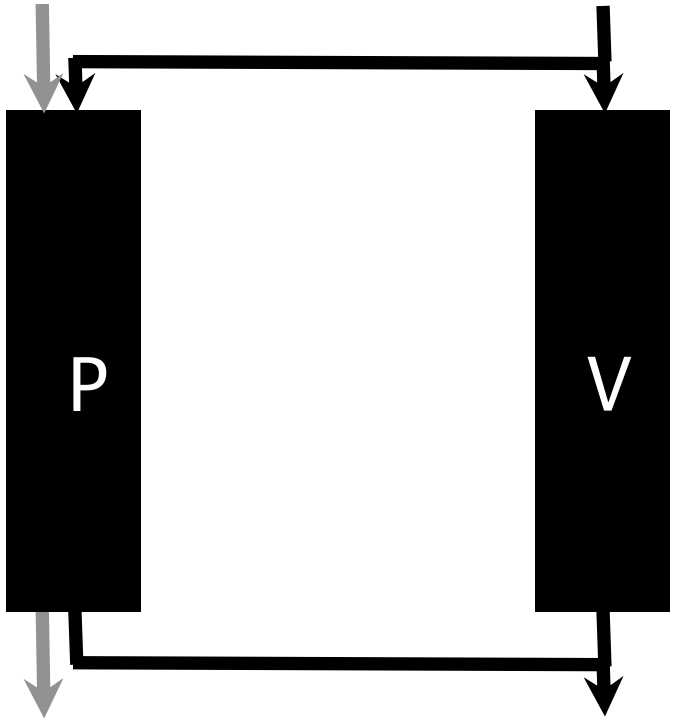
Problem: Verifier cannot enforce that prover folds W correctly

Folding Scheme for Relaxed R1CS from Additive Commitments

Solution: Treat (E, W) as the witness with commit in statement

(E_1, W_1) $(A, B, C, \overline{E}_1, \overline{W}_1, u_1, \mathrm{x}_1)$

(E_2, W_2) $(A, B, C, \overline{E}_2, \overline{W}_2, u_2, \mathrm{x}_2)$



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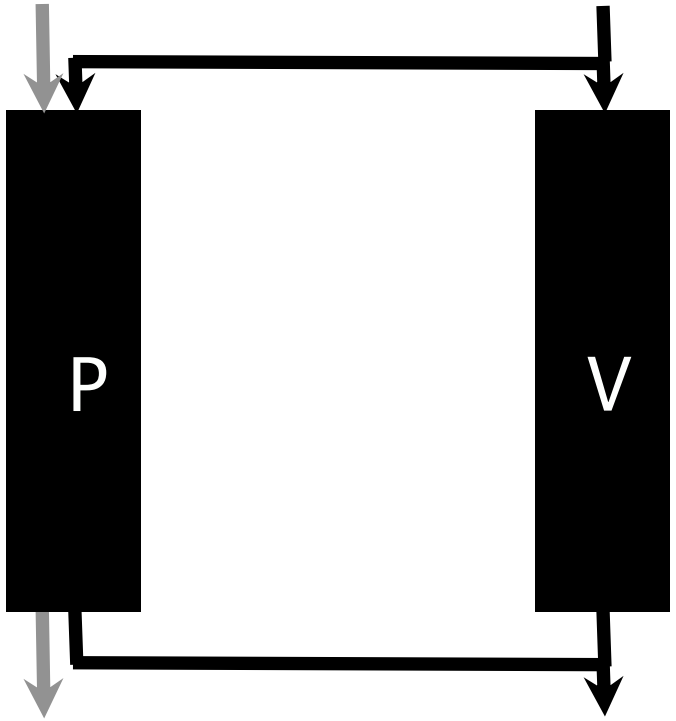

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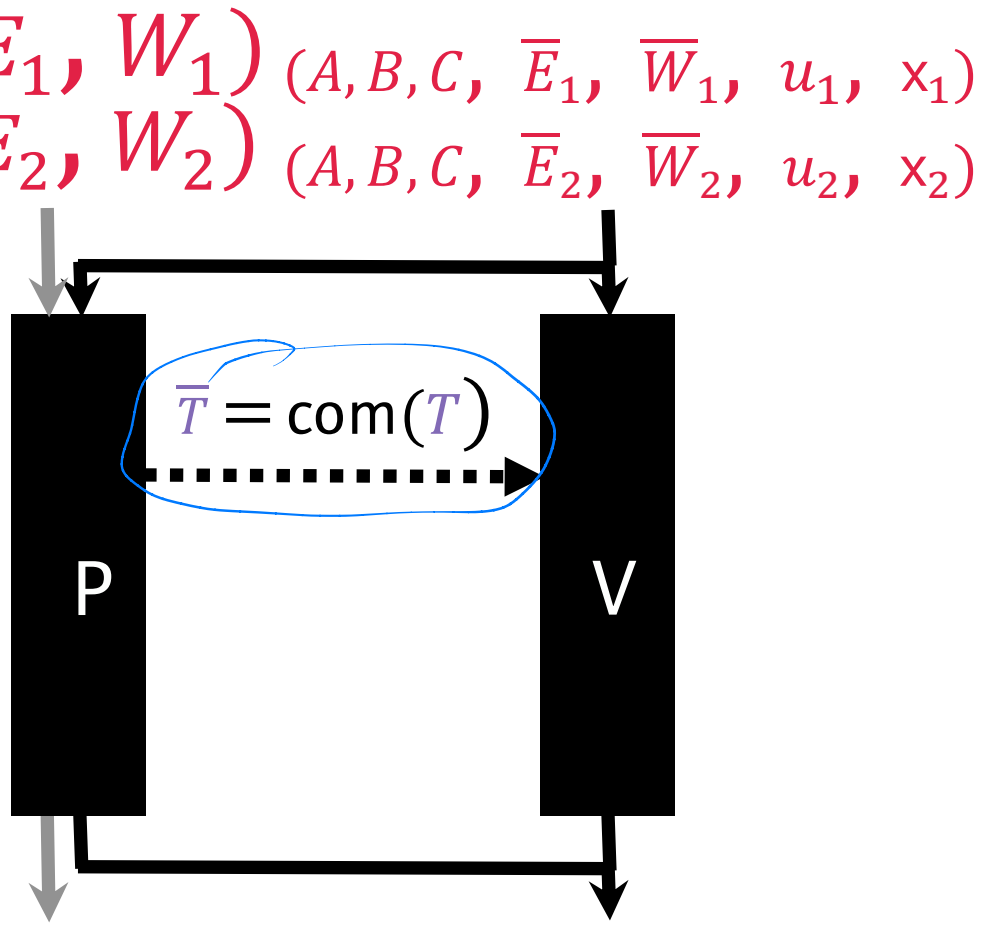

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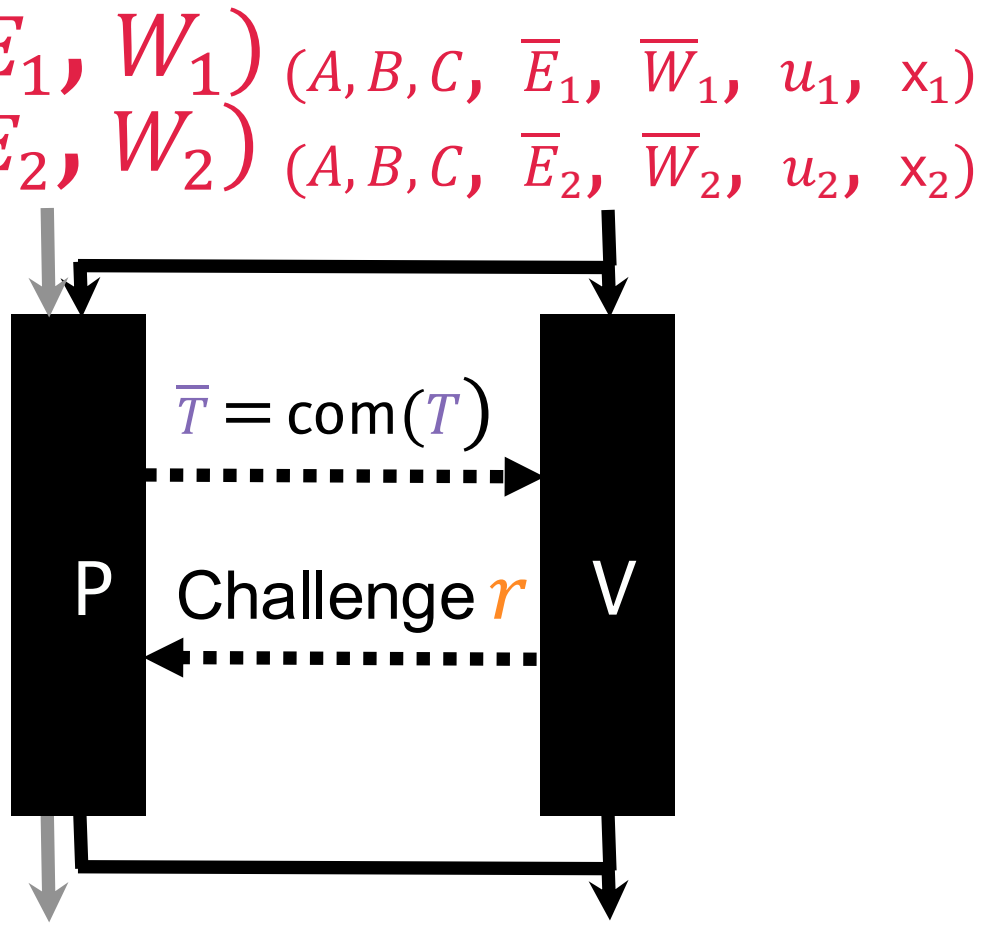

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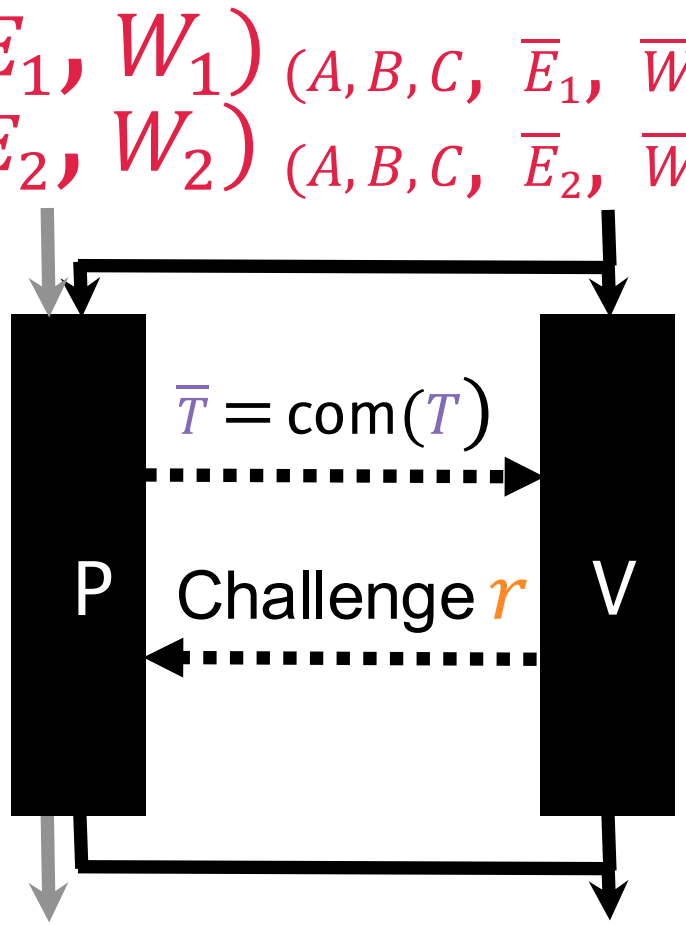

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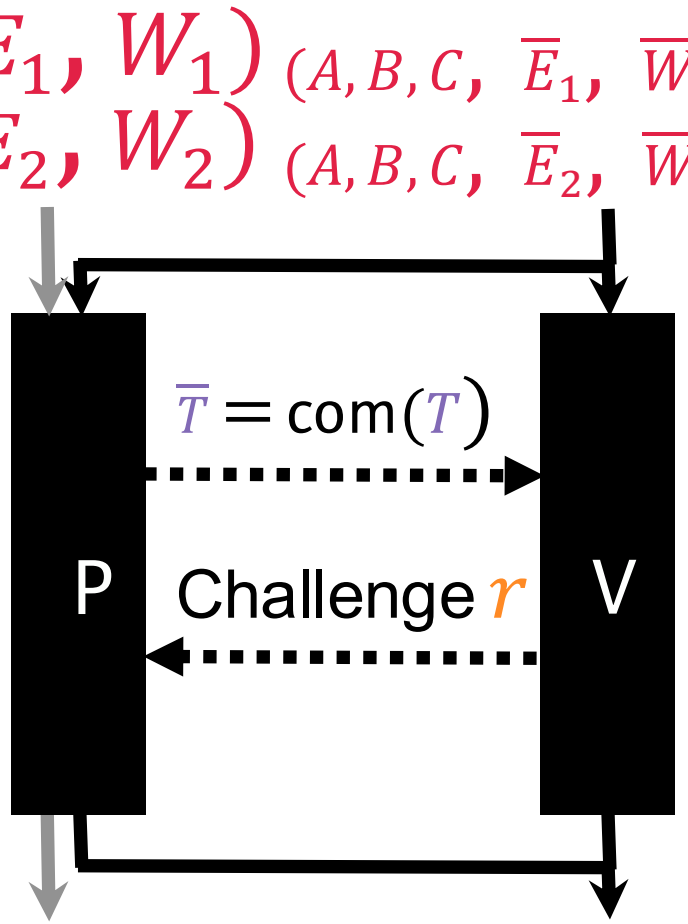

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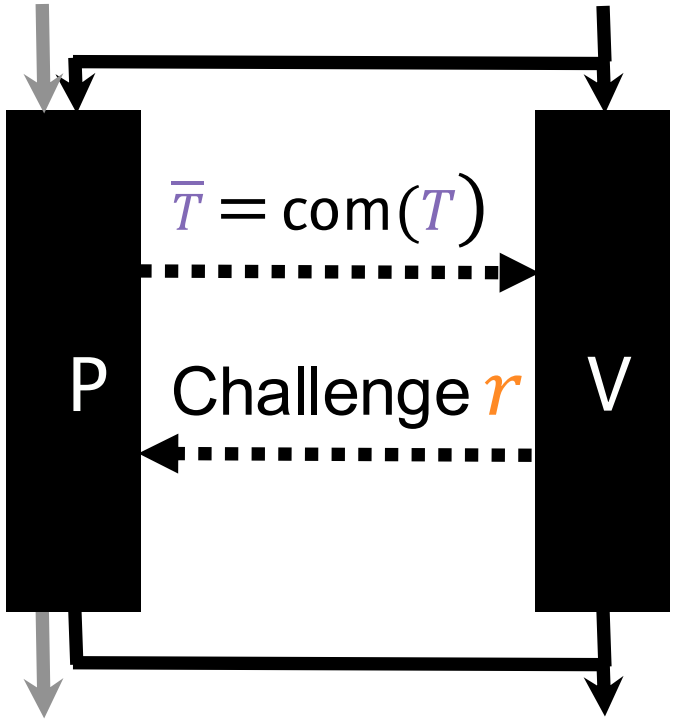

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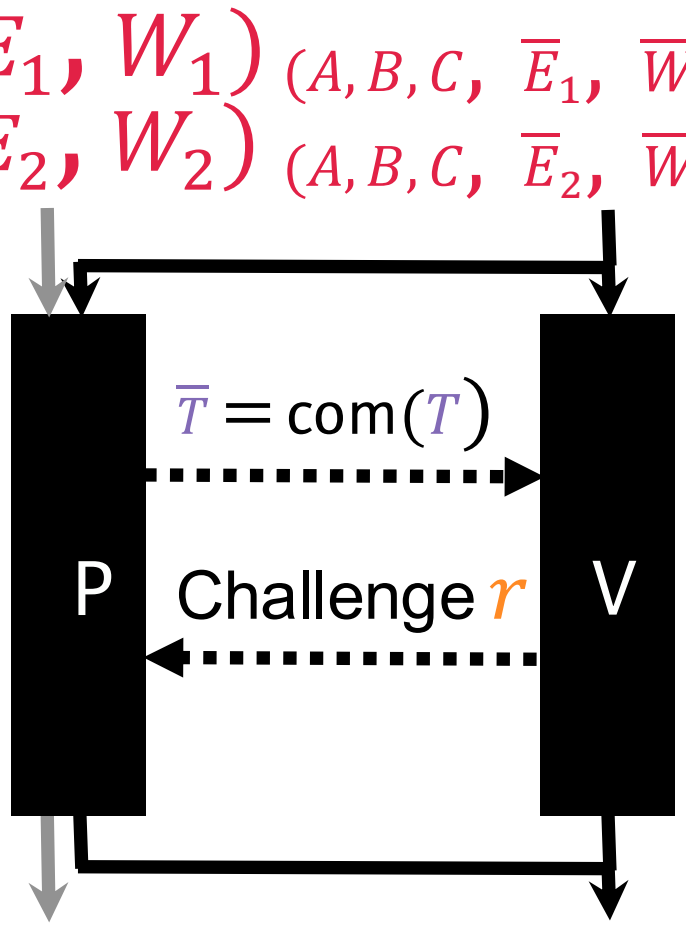

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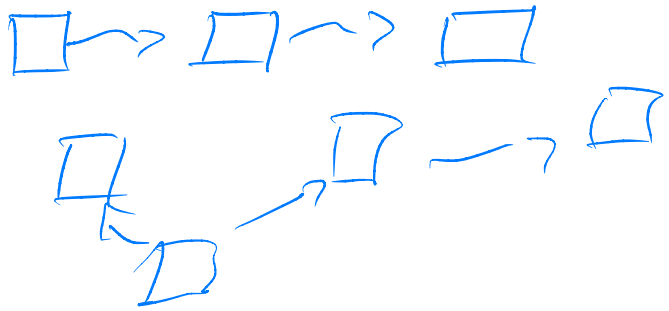
Agenda for this lecture

- Announcements
- Motivating incremental proofs
- Incrementally verifiable computation (IVC) via recursion
- IVC via folding, the Nova construction
- Proof-carrying data

Proof-carrying data (PCD)

(use recursion)

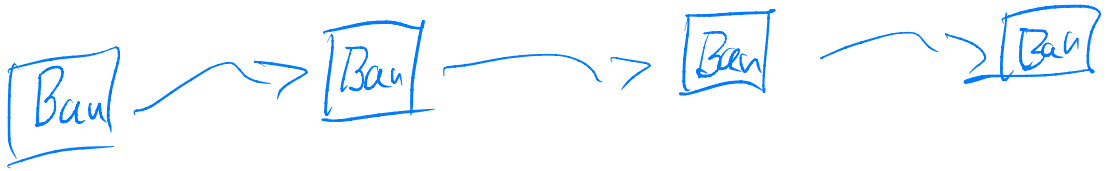
- Proof-carrying data
  IVC but a DAG



- non-uniform IVC
  Run different F functions

"switchboarding"

- Alpaca: anonymous moderation from IVC



- Post-quantum IVC